

Private Wealth & Portfolio Management: The Complete Course

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Win the case by making every holding trace back to the client's own words. This course teaches the full private-banking chain: understand the client, write the Investment Policy Statement (IPS), build the portfolio, manage its risk and defend it to judges.

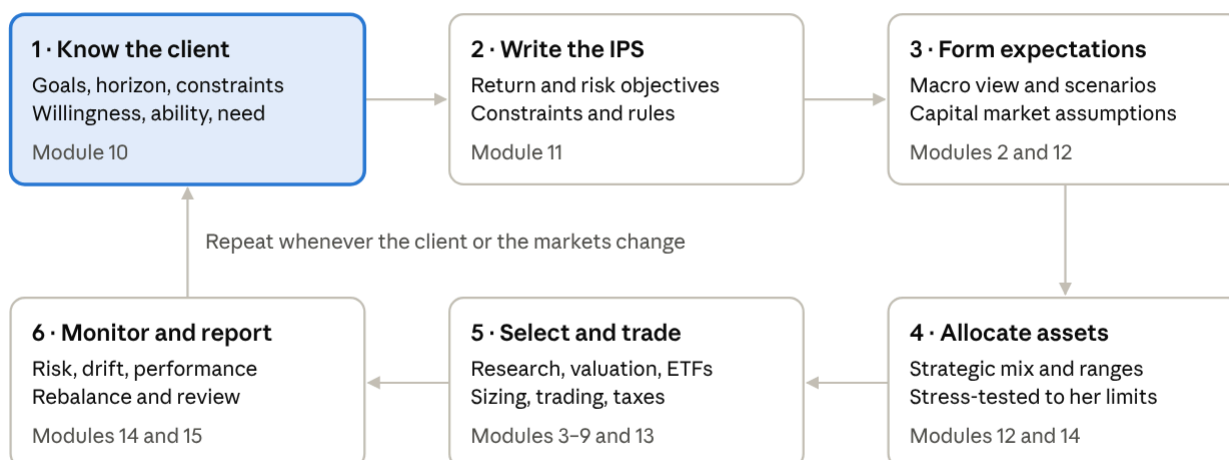
The one rule: fit beats return

In a case-based competition, a portfolio is judged on how well it serves the client's goals, risk tolerance and constraints, and on the reasoning behind it. Raw return matters far less than fit. For a conservative, risk-averse client like Laura Gao, capital preservation, inflation protection, liquidity and steady income usually outrank upside.

The portfolio management process

Private banks, endowments and the CFA curriculum all run the same six-step loop. Every module in this course plugs into one step of it.

Every decision starts with the client and loops back to her



The portfolio management process · 6 steps in a loop

Steps 1 and 2 decide what the portfolio must do, steps 3 to 5 build it, and step 6 checks it against the IPS.

Course map

Module	You will be able to	Feeds this part of your submission
1 · Investment foundations	Explain return, risk, compounding and every major asset class	Rationale, client education
2 · Economics for investors	Read the macro regime and link it to asset classes	Market outlook

Module	You will be able to	Feeds this part of your submission
3 · Business management and strategy	Judge business quality with Five Forces, moats and governance	Stock theses
4 · Financial statements and analytics	Compute and interpret ratios, spot accounting red flags	Stock theses
5 · Analyzing company growth	Measure growth and judge whether it creates value	Stock theses
6 · Equity valuation	Value a company with DCF, dividend discount and multiples	Buy and sell decisions
7 · Fixed income and TIPS	Analyze bonds, duration, credit and inflation protection	Bond and TIPS sleeve
8 · ETFs, funds and index investing	Choose ETFs on cost, tracking, liquidity and fit	Core holdings
9 · Alternatives and real assets	Decide whether REITs, gold or alternatives belong	Diversifiers
10 · Know your client	Turn the case file into goals and a risk profile	Client profile
11 · Building the IPS	Write a measurable, defensible IPS	IPS section
12 · Portfolio theory and asset allocation	Build and justify an asset allocation	Allocation section
13 · Construction and implementation	Pick, size, trade and rebalance holdings	Holdings, trade log
14 · Risk assessment and management	Measure, stress-test and manage risk	Risk section
15 · Performance and reporting	Measure results correctly and report them	Results, monitoring plan
16 · Competition playbook	Run the case workflow and answer judges	Report, presentation, Q&A
Appendix · Worked example	Follow one full case from profile to portfolio	A model to imitate

How to study it

- **Short on time?** Read Modules 10, 11, 12, 14 and 16 first. They are the critical path for the case.
- **Split by role.** Equity analysts take Modules 3 to 6, the fixed-income lead takes 7, and the ETF and risk lead takes 8, 9 and 14.
- **Keep the booklet open.** Every formula and a glossary of terms and acronyms live in the Formula & Key Terms Booklet.
- **Apply as you go.** Each module ends with "Apply it to your case". Do those steps on the Laura Gao file before moving on.

Conventions

- Figures labelled "approximate" are rounded long-run historical numbers for teaching. Check current data before you cite any number in your report.
- "Banker's note" callouts are practitioner tips from the private-banking desk.
- Tickers are examples to research, never recommendations. Verify fees and holdings on the issuer's website.

Module 1 · Investment foundations

Every investment decision trades off five things: return, risk, liquidity, time and taxes. From 1928 to 2023, US stocks compounded at 9.8% a year and 3-month T-bills at 3.3%. That gap is the reward for bearing risk, and this module shows how to measure both sides of it.

1.1 Time value of money

A dollar today is worth more than a dollar later because it can earn a return in the meantime. Bond pricing, valuation and retirement planning all come down to compounding forward or discounting back.

$$FV = PV \times (1 + r)^n \quad PV = \frac{FV}{(1 + r)^n}$$

- **Compounding:** \$100,000 invested at 5% for 10 years grows to \$162,889.
- **Rule of 72:** the years to double are roughly $72 \div$ the rate in percent. At 6% that gives 12 years; the exact answer is 11.9.
- **Inflation runs compounding in reverse:** after 10 years of 3% inflation, \$100,000 buys only what \$74,409 buys today.
- **Real vs nominal (Fisher):** the real return is what you earn in purchasing power.

$$1 + r_{real} = \frac{1 + r_{nominal}}{1 + \pi} \quad r_{real} \approx r_{nominal} - \pi$$

Example: a 6% nominal return with 3% inflation is a 2.91% real return (the shortcut gives 3%).

Banker's note: State a conservative client's return goal in real terms. A 4% return feels safe, but with 2.5% inflation it adds only about 1.5% a year of purchasing power.

1.2 Measuring return

- **Holding period return (HPR):** $(\text{ending price} - \text{beginning price} + \text{income}) \div \text{beginning price}$.
- **Total return:** price change plus income (dividends, interest), with the income reinvested. Always compare assets on total return.
- **Arithmetic vs geometric mean:** the arithmetic average of yearly returns overstates what an investor actually earns when returns swing. The geometric mean, also called the compound annual growth rate (CAGR), is the true growth rate. A gain of 50% followed by a loss of 50% averages 0%, but \$100 becomes \$150 and then \$75, a geometric return of -13.4% a year.
- **Volatility drag:** the geometric mean is roughly the arithmetic mean minus half the variance. For the S&P 500 from 1928 to 2023, 11.66% arithmetic minus half of 19.55% squared (1.91 points) is close to the actual 9.80% geometric return.
- **Annualizing:** $(1 + \text{cumulative return})$ raised to the power $1 \div \text{years}$, minus 1. Do not annualize periods shorter than a year in a client report, since it exaggerates.

- **Which return the client lives on:** nominal → after fees → after taxes → after inflation. Report the last one when you discuss goals.

1.3 Understanding risk

Risk is the chance that outcomes differ from what you expect: above all, the chance of losing money or missing a goal.

- **How to measure it:** standard deviation (volatility), maximum drawdown, worst year, probability of loss and probability of missing the goal (shortfall risk). Module 14 covers each.
- **Systematic vs unsystematic:** diversification removes company-specific (unsystematic) risk but not market-wide (systematic) risk. Markets pay a premium only for systematic risk (CAPM, Module 12).
- **The record:** US asset classes, 1928–2023, before fees and taxes ([Damodaran Online](#)):

Asset (US)	Compound annual return	Annual volatility	Worst year	Losing years (of 96)
S&P 500 stocks, dividends reinvested	9.80%	19.55%	−43.84% (1931)	26
Baa-rated corporate bonds	6.68%	7.71%	−15.68% (1931)	16
10-year Treasury bonds	4.57%	7.95%	−17.83% (2022)	19
3-month Treasury bills	3.30%	3.01%	+0.03% (2014)	0

Inflation averaged roughly 3% a year over this period (approximate), so T-bills barely kept pace while stocks multiplied purchasing power. Note 2022: 10-year Treasuries had their worst year in the dataset, losing almost as much as stocks (−17.8% vs −18.0%). Treasuries carry no default risk, but they carry plenty of interest-rate risk. Safe is not the same as riskless.

1.4 The asset classes

Asset class	What you own	Where the return comes from	Main risks	Role for a conservative client
Cash equivalents (T-bills, money market funds, CDs)	Very short-term loans to the government or banks	Interest	Inflation, falling rates on reinvestment	Liquidity reserve, near-term goals
Treasury bonds	Loans to the US government	Coupons and price changes	Interest-rate and inflation risk	Ballast in recessions and market crashes
TIPS and I bonds	Inflation-indexed government debt	Real yield plus a CPI adjustment	Real interest-rate risk	Protect purchasing power
Investment-grade corporate bonds	Loans to financially strong companies	Coupons above Treasuries (the credit spread)	Spread widening, interest rates	Extra income over Treasuries
High-yield (junk) bonds	Loans to weaker companies	High coupons	Default; behave like stocks in crises	Small or none
Municipal bonds	Loans to states and cities	Federally tax-exempt interest	Credit, interest rates	Income in high-bracket taxable accounts
US stocks	Ownership of US companies	Earnings growth, dividends, valuation changes	Bear markets, recessions	Long-term growth above inflation

Asset class	What you own	Where the return comes from	Main risks	Role for a conservative client
International stocks (developed and emerging)	Ownership of non-US companies	Same, plus currency moves	Currency, political risk	Diversification, cheaper valuations
Real estate (REITs)	Income-producing property	Rents and property values	Interest rates, leverage	Income, partial inflation hedge
Commodities and gold	Raw materials, precious metal	Price changes (futures roll for commodities)	High volatility, no income	Small inflation or crisis hedge
Private equity, private credit, hedge funds	Illiquid or complex strategies	Manager skill, illiquidity premium	Lock-ups, fees, opacity	Rarely; very wealthy clients only
Digital assets	Crypto tokens	Price changes only	Extreme volatility, regulation	Generally none

1.5 Investment vehicles

Vehicle	How it trades	Typical annual cost	Tax efficiency	Best use
Individual stocks	Intraday on exchanges	Near-zero commissions plus the bid-ask spread	High: you choose when to sell	Conviction ideas, satellite positions
Individual bonds	Over the counter through dealers	Dealer markup built into the price	You control timing	Known maturity dates, bond ladders
ETFs	Intraday on exchanges	From about 0.03% for broad index funds; more for niche funds	High, thanks to in-kind redemptions	Core building blocks
Mutual funds	Once a day at net asset value (NAV)	From about 0.03% (index) to 1%+ (active); some charge sales loads	Lower: can distribute capital gains	Retirement plans, active managers
Closed-end funds	Intraday, often at a premium or discount to NAV	Often 1%+ including leverage costs	Medium	Niche income strategies
Money market funds	At a stable \$1 NAV (government funds)	Low	Interest taxed as income	Cash management
Certificates of deposit (CDs)	Bank deposit; penalty for early withdrawal	No explicit fee	Interest taxed as income	FDIC-insured fixed rates
Annuities	Insurance contract	Near zero (simple fixed) to 2–3% (variable)	Tax-deferred growth	Guaranteed lifetime income (Module 9)

1.6 How markets work

- **Primary vs secondary market:** companies and governments raise money in the primary market (IPOs, new bond issues); investors then trade with each other in the secondary market.
- **Exchanges and market makers:** the NYSE and Nasdaq match buyers and sellers. Market makers quote a bid (what they pay) and an ask (what they charge); the gap is the bid-ask spread, a hidden trading cost.
- **Order types:** a market order fills now at whatever price is available. A limit order fills only at your price or better. A stop order becomes a market order once a trigger price trades; a stop-limit becomes a limit order. Use limit orders for ETFs and thinly traded stocks.
- **Settlement:** US stock and ETF trades settle one business day after the trade (T+1).

- **Market capitalization:** share price $\times$ shares outstanding. Common conventions put large caps above \$10 billion and small caps below \$2 billion.
- **Indexes:** the S&P 500 (500 large US companies, weighted by market value), the Dow Jones Industrial Average (30 stocks, weighted by share price), the Russell 2000 (US small caps), MSCI EAFE (developed markets outside North America), MSCI ACWI (global stocks) and the Bloomberg US Aggregate (US investment-grade bonds).
- **Margin and short selling:** borrowing to invest magnifies losses, and a short sale can lose more than 100%. Neither belongs in a conservative mandate.
- **Efficient market hypothesis (EMH):** prices reflect past prices (weak form), all public information (semi-strong) or all information (strong). Whichever form you believe, beating the market after costs is hard, so every active bet needs a reason.

1.7 The five levers you control

You cannot control returns. You can control these five, and judges reward teams that show they used them deliberately:

1. **Asset allocation:** the mix of asset classes (Module 12).
2. **Diversification:** across and within asset classes, regions and issuers.
3. **Costs:** fund fees, trading spreads and turnover.
4. **Taxes:** which account holds which asset, and how long you hold it (Module 13).
5. **Behavior:** a written plan, disciplined rebalancing and no panic selling.

Apply it to your case

- List each of Laura's goals with its horizon in years.
- Inflate each goal to future dollars: goal in today's dollars $\times (1 + \text{inflation})^{\text{years}}$.
- Record every liquidity need: amount and date.
- Name the risks she fears most: losing money, inflation, or running out of money.
- Draft one sentence to refine in Modules 10–11: "Laura needs a real return of about X% a year without a calendar-year loss larger than Y%."

Module 2 · Economics for investors

Macroeconomics sets the tide for every asset class. You do not need to forecast it precisely. You need to know which regime you are in, and to build a portfolio that survives the regimes you did not predict.

2.1 The indicators that matter

Indicator	What it measures	Released	How investors read it
Real GDP growth	Output of the economy after inflation	Quarterly (BEA)	Two negative quarters is a popular rule of thumb, but the NBER dates US recessions from a broader set of data
Unemployment rate and nonfarm payrolls	Labor market health	Monthly, usually the first Friday (BLS)	The Sahm rule flags recession when the 3-month average unemployment rate rises 0.5 points above its low of the prior 12 months
CPI and core CPI	Consumer prices; core excludes food and energy	Monthly (BLS)	Drives TIPS principal, Fed policy and bond yields
PCE price index	The inflation measure behind the Fed's 2% target	Monthly (BEA)	The Fed's preferred gauge
PPI	Prices producers receive	Monthly (BLS)	Early read on pipeline inflation and profit margins
ISM manufacturing and services PMI	Purchasing-manager survey; 50 separates expansion from contraction	Monthly	Leads the earnings cycle
Retail sales and consumer sentiment	Consumer spending, roughly two-thirds of US GDP	Monthly	Strength of demand
Initial jobless claims	New unemployment filings	Weekly (Department of Labor)	The fastest labor-market signal
Housing starts and permits	New construction	Monthly	A rate-sensitive leading sector
Yield curve (10-year minus 2-year, 10-year minus 3-month)	Long rates vs short rates	Daily (Treasury)	Inversion has preceded most US recessions
High-yield credit spreads	Extra yield on risky corporate debt	Daily	Widening signals financial stress
Leading Economic Index	Composite of leading indicators	Monthly (Conference Board)	Helps spot turning points

2.2 GDP and growth

$$GDP = C + I + G + (X - M)$$

- **Components:** consumption, investment, government spending and net exports (exports minus imports).
- **Nominal vs real:** nominal GDP is measured at current prices; real GDP strips out inflation. The GDP deflator is $\text{nominal} \div \text{real} \times 100$.

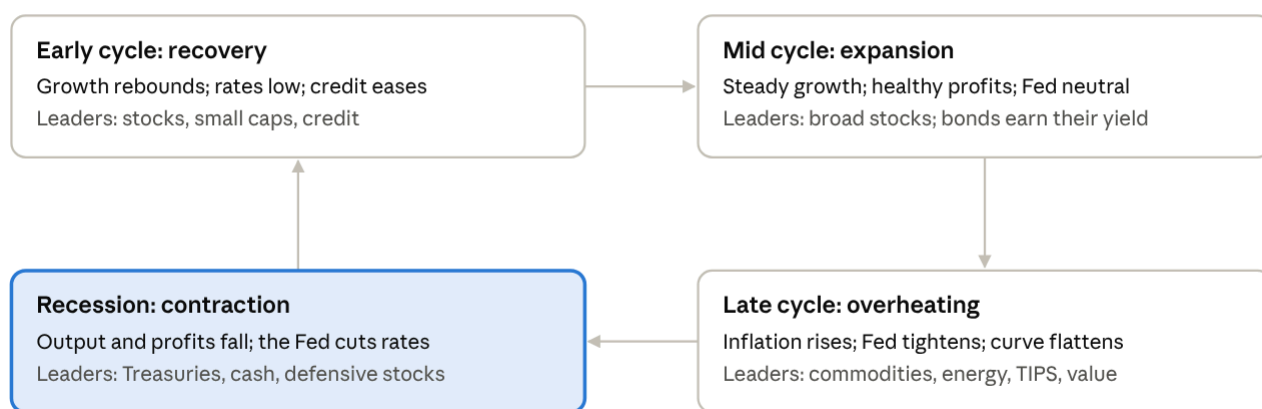
- **Potential output and the output gap:** when the economy runs above its potential, inflation tends to rise; when it runs below, there is slack and inflation tends to fall.
- **Long-run growth:** labor-force growth plus productivity growth, roughly 2% a year in real terms for the US (approximate). Long-run nominal GDP growth caps the terminal growth rate you can assume in a DCF (Module 6).

2.3 The business cycle

Economies move through recovery, expansion, overheating and recession. Each phase favors different assets, which is the logic behind sector rotation and tactical allocation.

Bonds and defensives lead in recessions; stocks and credit lead recoveries

Typical leadership by phase; real cycles vary in length and cause



The business cycle · 4 phases and the assets that tend to lead

Treat these as tendencies, not rules. Phases vary in length, and the drivers matter. In 2022 an inflation shock hit stocks and bonds at the same time, so the usual recession playbook of long bonds failed.

2.4 Inflation

- **Causes:** demand-pull (too much spending chasing too few goods), cost-push (energy or supply shocks), expectations that feed into wages, and monetary expansion ($MV = PY$).
- **Measures:** headline CPI, core CPI, PCE and core PCE. The Fed targets 2% on PCE.
- **Disinflation, deflation, stagflation:** disinflation is slowing inflation; deflation is falling prices; stagflation is high inflation with weak growth, the worst regime for a stock-and-bond portfolio.
- **Breakeven inflation:** a nominal Treasury yield minus the TIPS real yield of the same maturity. It is the market's inflation expectation plus risk premia (Module 7).

Asset	When inflation runs high or surprises upward	Why
Cash	Keeps up only as fast as short rates rise	Rates reset, but with a lag

Asset	When inflation runs high or surprises upward	Why
Nominal bonds	Hurt	Fixed payments lose value and yields rise
TIPS	Protected against realized inflation	Principal tracks CPI, but prices still fall if real yields rise
Stocks	Mixed	Firms with pricing power pass costs on; valuations fall as rates rise
Real estate	Partial hedge	Rents reset, but property is rate-sensitive
Commodities	Tend to benefit, especially from surprises	They are often the source of the inflation
Gold	Mixed in the short run	Tends to do well when real interest rates fall

2.5 Monetary policy and the Federal Reserve

- **Mandate:** maximum employment and stable prices (the dual mandate), with a 2% inflation goal.
- **The FOMC:** the 7 governors plus the New York Fed president and 4 rotating regional presidents vote. It meets 8 times a year and publishes the Summary of Economic Projections, including the "dot plot" of rate forecasts, at 4 of those meetings.
- **Tools:** the federal funds target range, interest on reserve balances, open market operations, the balance sheet (quantitative easing buys bonds to push long rates down; quantitative tightening shrinks holdings), the discount window and forward guidance.
- **Transmission:** the policy rate moves short-term rates, borrowing costs, asset prices, the dollar and expectations, which then move spending, hiring and inflation. The effects arrive with long and variable lags, often a year or more.
- **Reading the Fed:** "hawkish" leans toward higher rates to fight inflation; "dovish" leans toward lower rates to support growth. Fed funds futures (summarized by CME FedWatch) show the path the market expects.
- **Taylor rule:** a benchmark for where the policy rate "should" be, given inflation and the output gap (formula in the booklet).
- **Portfolio impact:** rising rates hurt long-duration assets (long bonds, high-valuation growth stocks); cuts help them. Cash yields follow the policy rate down as well as up.

2.6 Fiscal policy

- **Levers:** government spending and taxes. Deficits are financed by issuing Treasuries, which adds to the supply of bonds.
- **Multiplier and crowding out:** spending can lift GDP by more than its cost, but heavy borrowing can push up rates and crowd out private investment.
- **Automatic stabilizers:** unemployment insurance and progressive taxes cushion recessions without new legislation.
- **Market impact:** large, persistent deficits can raise long-term yields through the term premium. All three major agencies now rate US debt one notch below AAA (S&P in 2011, Fitch in 2023, Moody's in 2025).

2.7 Interest rates and the yield curve

- **Shapes:** normal (upward sloping), flat, inverted (short rates above long rates) and humped.
- **Why it slopes:** expectations theory says long rates average the expected short rates; liquidity preference adds a term premium for lending longer; preferred habitat says different investors favor different maturities.
- **As a signal:** inversions have preceded most US recessions of the past 60 years, with lags of roughly 6 to 24 months. The long 2022–24 inversion was not followed by a recession within that window, a reminder that it is a signal, not a schedule.
- **Real vs nominal:** nominal yield $\approx$ real yield + expected inflation + inflation risk premium. Treasury publishes both nominal and TIPS real yield curves daily.

2.8 Global macro and currencies

- **What moves exchange rates:** interest-rate differentials, inflation differentials (purchasing power parity), relative growth, risk appetite, trade balances and capital flows.
- **Your return in dollars:** $(1 + \text{local return}) \times (1 + \text{currency return}) - 1$. A 10% gain in European stocks becomes about 4.5% if the euro falls 5% against the dollar.
- **Hedging:** currency swings dominate the volatility of foreign bonds, so foreign bonds are usually held currency-hedged. For foreign stocks, leaving currency unhedged is common because it adds diversification.
- **Emerging markets:** faster growth, but more political, currency, governance and liquidity risk.
- **Global shocks:** tariffs and trade policy, geopolitics, oil supply decisions, China's growth, and other central banks (ECB, Bank of Japan, Bank of England).
- **The dollar:** as the world's reserve currency, it has usually risen in global crises, which cushions US investors holding foreign assets.

2.9 Microeconomics for analysts

- **Elasticity:** if demand barely falls when price rises (inelastic demand), the company has pricing power. That is the single most useful micro idea for stock analysis.
- **Costs:** fixed vs variable costs determine operating leverage, meaning how much profit swings when sales swing. Profit is maximized where marginal revenue equals marginal cost.
- **Scale and networks:** economies of scale lower unit costs as volume grows; network effects make a product more valuable as more people use it.
- **Opportunity cost:** every investment competes with the next-best alternative. For a conservative client, that hurdle is the T-bill rate.
- **Game theory:** price wars are a prisoner's dilemma; industries with a few rational players tend to avoid them.

Market structure	Number of firms	Pricing power	Typical margins	Examples
Perfect competition	Many, identical products	None (price takers)	Thin	Grain farming

Market structure	Number of firms	Pricing power	Typical margins	Examples
Monopolistic competition	Many, differentiated products	Some, via brand	Moderate	Restaurants, apparel
Oligopoly	A few, interdependent	Significant	Healthy	Wireless carriers, card networks
Monopoly	One	High, often regulated	High or capped by regulators	Regulated utilities

2.10 Building your macro view

Write the view in five lines, then turn it into scenarios:

1. **Growth:** where are we in the cycle, and what do the leading indicators say?
2. **Inflation:** the trend, and what breakevens expect.
3. **Policy:** the Fed's projected path vs what futures markets price in.
4. **Valuations:** stock multiples vs history, bond yields vs inflation, credit spreads vs average.
5. **Risks:** what would break the base case?

Scenario	What happens	Typical effect on assets	What protects the client
Soft landing	Growth slows but stays positive; inflation drifts toward target	Stocks and bonds both earn positive returns	The core allocation
Recession	Unemployment rises, profits fall, the Fed cuts	Stocks fall; Treasuries and cash gain	High-quality bonds, cash reserve
Inflation resurgence	Inflation re-accelerates; rates rise	Stocks and bonds fall together, as in 2022	Short duration, T-bills, TIPS, real assets
Productivity boom	Strong growth with tame inflation	Stocks rally; bonds flat	Equity sleeve, rebalancing discipline

Assign your own probabilities, then check that the portfolio is acceptable in every row, not just the likeliest one.

2.11 Where to get the data

- [FRED](#) (St. Louis Fed): nearly every US and many global series, free and downloadable.
- BLS (CPI, jobs), BEA (GDP, PCE) and the Federal Reserve (FOMC statements, minutes, projections).
- US Treasury: daily nominal and real (TIPS) yield curves.
- CME FedWatch: market-implied odds of Fed moves.
- J.P. Morgan Guide to the Markets: a quarterly chart book, ideal for a market-outlook section.

Apply it to your case

- Write your five-line macro view and one scenario table for the report.
- For each scenario, estimate Laura's portfolio return (Module 14 shows how).
- Make sure the portfolio survives the scenario that is least likely but most damaging. For a conservative client that is usually a 2022-style inflation shock or a 2008-style crash.

Module 3 · Business management and strategy

Before you value a company, decide whether it is a good business: one that earns returns above its cost of capital and can defend them for years. For a conservative client, business quality is the first line of risk management, because great businesses recover from bad markets and weak ones often do not.

3.1 What makes a high-quality business

The thresholds below are rules of thumb; compare every company with its own industry.

Trait	What to look for	Where to check
High returns on capital	Return on invested capital (ROIC) above the cost of capital, for example 12%+ against 8%, sustained for many years	10-K history, data providers
Pricing power	Gross margins stable or rising, even through inflation	Gross margin trend
Recurring revenue	Subscriptions, long contracts, consumables	Revenue breakdown in the 10-K
Low capital intensity	Capex small relative to sales; most profit converts to free cash flow	Cash flow statement
Strong balance sheet	Net debt below about 2× EBITDA; operating profit covers interest 8× or more	Balance sheet, debt footnotes
Durable demand	Sales held up in 2008–09 and 2020	Segment results in those years
Aligned management	Meaningful insider ownership; pay tied to per-share value and returns	Proxy statement (DEF 14A)

3.2 Business models

Model	How it makes money	Key metrics	Watch out for
Subscription and software (SaaS)	Recurring fees	Annual recurring revenue, net revenue retention, churn, customer acquisition cost payback	High acquisition costs; dilution from stock-based pay
Marketplace or platform	A cut (take rate) of transactions	Gross merchandise value, take rate, active users	Regulation, being bypassed
Razor and blade	Cheap device, profitable refills	Installed base, attach rate	Cheaper copycat refills
Franchise	Royalties from franchisees	System-wide sales, unit growth, same-store sales	Franchisee financial health
Advertising	Sells audience attention	Users, engagement, revenue per user	Ad cycles, privacy rules
Licensing and royalties	Fees for intellectual property	Royalty rates, renewals	Patent expiry
Asset-heavy manufacturing	Sells physical goods	Capacity utilization, margins, inventory turns	Cyclical demand, heavy capex
Commodity producer	Sells at market prices	Cost per unit vs price, reserves	Price swings it cannot control
Regulated utility	An allowed return on its asset base	Rate-base growth, allowed return on equity	Regulators, interest rates

Model	How it makes money	Key metrics	Watch out for
Bank or insurer	The spread on deposits, or float on premiums	Net interest margin, combined ratio, capital ratios	Credit losses, leverage

3.3 Strategy frameworks

Porter's Five Forces tells you whether an industry lets companies keep their profits.

Force	Question to ask	Favorable sign for investors
Rivalry among competitors	Are price wars common?	A few rational competitors in a growing market
Threat of new entrants	How hard is it to start a rival?	High capital needs, regulation, brand, scale
Buyer power	Can customers force prices down?	Many small customers with high switching costs
Supplier power	Can suppliers squeeze margins?	Many suppliers of commodity inputs
Threat of substitutes	Could a different product do the same job?	No close substitute

Other frameworks judges expect you to know:

- **SWOT:** internal strengths and weaknesses; external opportunities and threats.
- **PESTEL:** political, economic, social, technological, environmental and legal forces on an industry.
- **Porter's generic strategies:** cost leadership, differentiation, or focus on a niche. Companies "stuck in the middle" tend to earn poor returns.
- **Value chain:** primary activities (inbound logistics, operations, outbound logistics, marketing and sales, service) plus support activities. Find where the margin is created.
- **VRIO:** a resource gives lasting advantage if it is Valuable, Rare, hard to Imitate, and the firm is Organized to exploit it.
- **BCG growth-share matrix:** stars, cash cows, question marks and dogs, for companies with several segments.
- **Ansoff matrix:** market penetration, market development, product development and diversification, in rising order of risk.
- **Industry life cycle:** introduction, growth, shakeout, maturity, decline. Mature industries often suit income-focused clients.
- **Balanced Scorecard:** financial, customer, internal-process and learning-and-growth measures of performance.
- **Marketing mix (4Ps):** product, price, place and promotion.

3.4 Economic moats

A moat is a durable advantage that keeps competitors from eroding returns. Morningstar's framework names five sources:

Moat source	How it works	Typical examples
Intangible assets	Brands, patents and licenses that competitors cannot copy	Consumer brands, drug patents
Switching costs	Leaving is costly, risky or time-consuming for customers	Enterprise software, banking relationships
Network effects	Each new user makes the product more valuable	Payment networks, marketplaces

Moat source	How it works	Typical examples
Cost advantage	Structurally lower costs from scale, process or location	Discount retailers, low-cost producers
Efficient scale	A market only big enough for one or a few players	Pipelines, airports, railroads

Test a claimed moat with numbers: ROIC above the cost of capital for 10+ years, stable or rising market share, margins that held up in downturns, and price increases at or above inflation without losing customers. Moats erode through technology shifts and regulation, so ask what could breach this one.

3.5 Management and capital allocation

Management has five uses for cash: reinvest in the business, acquire companies, pay dividends, buy back shares or repay debt. Good capital allocators pick the option with the highest risk-adjusted return per dollar and explain why.

- **Judge the record:** compare past promises with results; read several years of shareholder letters and earnings calls.
- **Check alignment:** insider ownership, and pay tied to ROIC or per-share free cash flow rather than revenue or "adjusted" EPS.
- **Buybacks:** create value only when shares are bought below intrinsic value. Check whether the share count actually falls, or whether buybacks merely offset stock-based compensation.
- **Red flags:** serial acquisitions followed by goodwill write-downs; debt-funded buybacks at peak prices; "adjusted" metrics that exclude recurring costs; departures of the CFO or auditor; related-party deals; guidance that is always beaten by a cent.

3.6 Corporate governance

- **The agency problem:** managers (agents) may act in their own interest rather than shareholders' (principals). Governance exists to align them.
- **Board quality:** a majority of independent directors, an independent chair or lead director, relevant expertise and active audit, compensation and nominating committees.
- **Shareholder rights:** one share, one vote is best. Dual-class shares, staggered boards and poison pills entrench management.
- **Audit and controls:** a reputable auditor, a clean opinion, no material weaknesses, no restatements.
- **Where to look:** the proxy statement (DEF 14A) for pay and the board, the 10-K for risk factors and internal controls, and 8-K filings for sudden executive departures.

3.7 ESG and sustainable investing

ESG covers environmental (climate, emissions, resource use), social (labor, product safety, data privacy, communities) and governance (board, pay, shareholder rights, ethics) issues. Focus on issues that are financially material in that industry; the SASB standards, now maintained by the ISSB, map them industry by industry.

Approach	What it does	Example
Exclusionary screening	Removes industries or companies	No tobacco or weapons makers
ESG integration	Adds ESG risks to the financial analysis	Higher discount rate for poor governance
Best-in-class	Picks the ESG leaders within each sector	Top-rated utilities only
Thematic	Invests in a sustainability trend	Clean energy, water
Impact investing	Targets measurable social or environmental outcomes	Green bonds, community lending
Stewardship	Votes proxies and engages management	Pushing for an independent chair

Pitfalls: ESG ratings from different providers often disagree, "greenwashing" overstates credentials, and exclusions shift sector weights away from the benchmark. In a competition, use ESG when the client states values or when an issue is financially material, and record it in the IPS under unique circumstances.

3.8 Corporate finance: how managers make decisions

- **Capital budgeting:** accept projects with a positive net present value (NPV) at a discount rate that matches their risk. The internal rate of return (IRR) must beat the hurdle rate, but beware projects of different sizes and cash flows that change sign more than once. Payback period is simple but ignores later cash flows.
- **Cost of capital:** the weighted average cost of capital (WACC) is the minimum return investors require (Module 6).
- **Capital structure:** in a frictionless market, Modigliani–Miller says the debt-equity mix does not change firm value. In reality, trade-off theory balances the tax shield on interest against the costs of financial distress, and pecking-order theory says firms use internal cash first, then debt, then new equity.
- **Payout policy:** dividends are sticky and signal confidence; buybacks are flexible. Check that free cash flow covers the dividend.
- **Mergers and acquisitions:** acquirers usually pay a premium over market price, so value depends on synergies outweighing that premium and integration going smoothly. Many deals fail that test, so treat a big acquisition as a risk until the results prove otherwise.
- **Working capital:** the cash conversion cycle (days of receivables + days of inventory – days of payables) shows how long cash is tied up in operations.
- **Operating leverage:** the degree of operating leverage (percent change in EBIT ÷ percent change in sales) shows how much profits swing when sales swing.

3.9 Operations and performance management

- **KPIs and OKRs:** good companies report the same key performance indicators every year and explain misses. Changing metrics often is a warning sign.
- **Unit economics:** revenue and cost per unit or per customer; the lifetime value of a customer should comfortably exceed the cost of acquiring one.

- **Supply chain:** check for single-source suppliers, geographic concentration and the inventory strategy (just-in-time is efficient but fragile; just-in-case is resilient but ties up cash).
- **Execution:** compare management's targets with what was delivered, year after year.

3.10 Qualitative research checklist

1. How does the company make money, in one sentence?
2. Who are its customers, and why do they choose it?
3. What is the moat, and is it widening or narrowing?
4. How do the Five Forces look for its industry?
5. How did revenue and margins hold up in 2008–09 and 2020?
6. Does management allocate capital well, and does it own shares?
7. Is governance sound: board, auditor, shareholder rights?
8. What are the top three risks in the 10-K, and how likely are they?
9. Which ESG issues are material in this industry?
10. Why does this business belong in this client's portfolio?

Apply it to your case

- For each stock candidate, write a one-paragraph business-quality summary covering the moat, the Five Forces and management.
- For a risk-averse client like Laura, favor wide-moat, low-debt, cash-generative businesses with steady dividends. Screen out companies that need everything to go right to work.

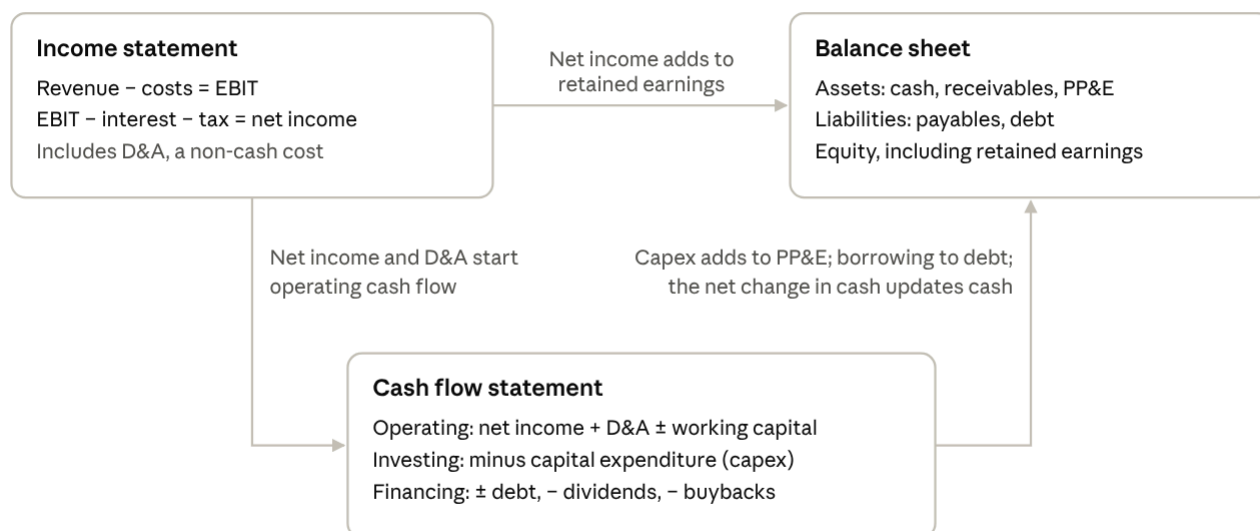
Module 4 · Financial statements and business analytics

The three financial statements show how a company makes money, what it owns and owes, and whether its profits turn into cash. Ratios turn them into signals you can compare across years and against peers.

4.1 The three statements and how they connect

- **Income statement (over a period):** revenue – cost of goods sold = gross profit; – operating expenses (selling, general and administrative; research and development) = operating income (EBIT); – interest = pretax income; – taxes = net income. EBITDA adds back depreciation and amortization.
- **Balance sheet (at a point in time):** assets = liabilities + shareholders' equity. Working capital = current assets – current liabilities.
- **Cash flow statement (over a period):** operating cash flow (CFO) starts from net income, adds back non-cash charges and adjusts for working capital; investing cash flow (CFI) covers capex and acquisitions; financing cash flow (CFF) covers debt, dividends, buybacks and share issuance.

Net income and cash tie the three statements together



How the three financial statements link · 3 statements, 3 links

If you can trace a change in one statement to the other two, you understand the business mechanically. That is the test analysts use in interviews, and it is how you catch numbers that do not add up.

4.2 Where to find the data

- **Filings, free on SEC EDGAR:** the 10-K (audited annual report), 10-Q (quarterly), 8-K (material events), DEF 14A (proxy: pay and board), S-1 (IPO registration) and 20-F (foreign companies).
- **Company sources:** earnings releases, call transcripts and investor-day presentations on the investor-relations site.
- **Read a 10-K in this order:** business description → risk factors → management's discussion and analysis (MD&A) → financial statements → footnotes (revenue recognition, debt maturities, leases, segments, stock compensation, contingencies).

4.3 The ratio toolkit

Rules of thumb vary by industry; always compare with peers and the company's own history. Full formulas are in the booklet.

Category	Ratio	Formula	What it tells you
Profitability	Gross margin	Gross profit ÷ revenue	Pricing power and production cost
Profitability	Operating margin	EBIT ÷ revenue	Core profitability before financing
Profitability	Return on assets (ROA)	Net income ÷ average total assets	How well assets produce profit
Profitability	Return on equity (ROE)	Net income ÷ average equity	Shareholder return; inflated by leverage
Profitability	Return on invested capital (ROIC)	After-tax operating profit (NOPAT) ÷ (debt + equity – cash)	The best single quality measure; compare with WACC
Efficiency	Asset turnover	Revenue ÷ average assets	Sales generated per dollar of assets
Efficiency	Days sales outstanding (DSO)	Receivables ÷ revenue × 365	How fast customers pay
Efficiency	Days inventory outstanding (DIO)	Inventory ÷ cost of goods sold × 365	How long stock sits on shelves
Efficiency	Days payables outstanding (DPO)	Payables ÷ cost of goods sold × 365	How long the firm takes to pay suppliers
Efficiency	Cash conversion cycle	DSO + DIO – DPO	Days cash is tied up in operations
Liquidity	Current ratio	Current assets ÷ current liabilities	Short-term bills covered; above 1 is typical
Liquidity	Quick ratio	(Cash + securities + receivables) ÷ current liabilities	Same, without inventory
Leverage	Net debt to EBITDA	(Debt – cash) ÷ EBITDA	Years of profit to repay debt; under 2–3× is comfortable for most non-financials
Leverage	Interest coverage	EBIT ÷ interest expense	Safety of interest payments; above 5× is comfortable
Cash flow	Free cash flow (FCF)	Operating cash flow – capex	Cash left for owners
Cash flow	FCF conversion	FCF ÷ net income	Earnings quality; near 100% is healthy
Payout	Payout ratio	Dividends ÷ net income (or ÷ FCF)	Dividend safety; lower leaves more cushion

4.4 DuPont analysis: where ROE really comes from

$$ROE = \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}$$

Company A earns a 20% net margin with asset turnover of 0.8 and leverage of 1.25: ROE = 20%. Company B earns a 5% margin with turnover of 1.0 and leverage of 4.0: ROE is also 20%. The same ROE hides very different risk. B depends on borrowed money, so a small fall in margins hits its equity far harder. A conservative client should prefer A.

The five-step version splits the margin further: ROE = tax burden (net income ÷ pretax income) × interest burden (pretax income ÷ EBIT) × operating margin × asset turnover × leverage.

4.5 Earnings quality and red flags

Red flag	What it may mean	How to check
Net income growing much faster than operating cash flow	Profits not turning into cash	CFO ÷ net income over 3–5 years
Receivables growing faster than revenue	Pushing product onto customers, loose credit	DSO trend
Inventory growing faster than sales	Weak demand, obsolete stock	DIO trend
Capitalizing costs that peers expense	Profits flattered	Accounting-policy footnote
"One-time" charges every year	Recurring costs hidden as unusual	Restructuring history
Large gap between GAAP and adjusted earnings	Heavy exclusions such as stock-based pay	Non-GAAP reconciliation
Goodwill large relative to equity	Overpaid acquisitions; write-down risk	Goodwill ÷ equity
Off-balance-sheet obligations	Hidden leverage	Commitments and contingencies note
Auditor change, late filing or restatement	Weak controls	8-K and late-filing notices
Clusters of insider selling	Insiders see limited upside	Form 4 filings

Three scoring models give a quick screen (formulas in the booklet): the Altman Z-score for bankruptcy risk, the Piotroski F-score for financial strength, and the Beneish M-score for possible earnings manipulation.

4.6 Industry KPIs

Industry	Metrics that matter
Software (SaaS)	Annual recurring revenue, net revenue retention, customer acquisition cost payback, Rule of 40 (growth % + FCF margin % of 40 or more)
Retail	Same-store sales, sales per square foot, inventory turns, online share
Restaurants	Same-store sales split into traffic and average ticket, unit growth, restaurant-level margin
Banks	Net interest margin, efficiency ratio, CET1 capital ratio, non-performing loans, loan-to-deposit ratio
Insurance	Combined ratio (below 100% means an underwriting profit), investment yield, reserve development
REITs	Funds from operations (FFO) and adjusted FFO per share, occupancy, same-store net operating income, debt to EBITDA
Oil and gas	Production, reserve life, breakeven oil price, reinvestment rate

Industry	Metrics that matter
Utilities	Rate-base growth, allowed return on equity, regulatory lag
Airlines	Load factor, revenue and cost per available seat mile
Semiconductors	Book-to-bill ratio, gross margin, factory utilization
Telecom	Average revenue per user, churn, subscriber additions
Consumer staples	Organic growth split into volume and price/mix, market share
Pharma and biotech	Pipeline stages, patent expiries, R&D productivity

4.7 Business analytics toolkit

- **Four levels of analytics:** descriptive (what happened), diagnostic (why), predictive (what will happen) and prescriptive (what to do about it).
- **Techniques:** horizontal analysis (trends over time), vertical or common-size analysis (each line as a percentage of revenue or assets), peer comparison tables, regression (beta, factor exposures), scenario and sensitivity analysis, Monte Carlo simulation and cohort analysis.
- **Statistics you will use:** mean, median, standard deviation, covariance, correlation (−1 to +1), regression slope and R^2 , t-statistics, the normal distribution (68–95–99.7 rule), skewness (lopsided returns) and kurtosis (fat tails).
- **Biases to avoid:** survivorship bias (failed funds vanish from databases), look-ahead bias (using data not available at the time), data mining (finding patterns in noise), tiny samples, and mistaking correlation for causation.
- **Spreadsheet skills:** XLOOKUP or INDEX-MATCH, SUMIFS, pivot tables, STDEV.S, CORREL, SLOPE (for beta), XIRR, XNPV, data tables for sensitivity, Goal Seek and Solver. Google Sheets' GOOGLEFINANCE function pulls prices. The booklet's section K lists them all.
- **Free data sources:** SEC EDGAR, company investor-relations sites, FRED, Yahoo Finance, stockanalysis.com, Macrotrends, Koyfin (free tier), the Finviz screener and Portfolio Visualizer (backtests, Monte Carlo, efficient frontiers).

4.8 One-page company tear sheet

Use the same template for every stock so the team can compare candidates side by side.

Section	What to fill in
Snapshot	Name, ticker, sector, market cap, what it does in one sentence
Five-year trend	Revenue, operating margin, EPS, free cash flow, ROIC
Balance sheet	Net debt to EBITDA, interest coverage, credit rating
Shareholder returns	Dividend yield, dividend growth, payout ratio, net buybacks
Valuation	P/E, EV/EBITDA and FCF yield vs the 5-year average and peers
Quality	Moat, Five Forces verdict, management and governance notes
Risks	Top three risks and what would make us sell
Verdict	Buy, hold or avoid; target weight; role in the client's portfolio

Apply it to your case

- Build a tear sheet for each stock candidate and put them in your report's appendix.
- Give extra scrutiny to any company with net debt above $3\times$ EBITDA or free-cash-flow conversion below 80%. A conservative client should not carry balance-sheet risk.

Module 5 · Analyzing company growth

Growth creates value only when a company earns more on new capital than that capital costs. So measure growth carefully, test its quality and durability, then ask how much of it the price already assumes.

5.1 Measuring growth

$$CAGR = \left(\frac{\text{Ending value}}{\text{Beginning value}} \right)^{1/n} - 1$$

- **Year-over-year (YoY):** this period ÷ the same period a year earlier – 1. Comparing a quarter with the same quarter last year removes seasonality.
- **CAGR:** EPS rising from \$2.00 to \$3.20 over 4 years is 12.5% a year.
- **Organic vs acquired:** strip out acquisitions and divestitures to see the underlying business.
- **Constant currency:** removes exchange-rate effects for multinationals.
- **Per share vs total:** net income can grow while EPS does not if the company issues shares. Owners care about per-share growth.
- **Every line:** measure growth in revenue, gross profit, EBIT, EPS, free cash flow, dividends and book value. When the lines diverge, find out why.

5.2 Worked example: Harbor Foods (fictional)

Harbor Foods has 150 million shares throughout; dollar figures are in millions except EPS.

Measure	Year 1	Year 2	Year 3	Year 4	Year 5	4-year CAGR
Revenue (\$M)	4,000	4,240	4,452	4,719	5,000	5.7%
Revenue growth		6.0%	5.0%	6.0%	6.0%	
Operating margin	12.0%	12.4%	12.9%	13.3%	13.8%	
EBIT (\$M)	480	526	574	628	690	9.5%
EPS (\$)	2.40	2.65	2.90	3.18	3.50	9.9%
Free cash flow (\$M)	380	410	430	470	520	8.2%
FCF ÷ net income	106%	103%	99%	99%	99%	

- **Operating leverage** turned 5.7% revenue growth into 9.5% EBIT growth: margins widened 1.8 points as fixed costs were spread over more sales.
- **EPS grew slightly faster than EBIT** because interest costs fell as debt was repaid; the share count was flat.
- **Cash backs the earnings:** free cash flow tracked net income (conversion near 100%), so the growth is real.
- **The valuation question:** can margins keep expanding? If they plateau near 14%, EPS growth slows toward revenue growth of about 6%. Never extrapolate 10% forever.

5.3 Quality of growth: eight tests

Test	Good sign	Warning sign
Organic	Most growth comes from the existing business	Growth mostly bought through acquisitions
Profitable	Margins stable or rising	Revenue up, margins down: growth bought with price cuts
Cash-backed	Free cash flow grows with earnings	Earnings up, cash flat as receivables or inventory build
Capital-efficient	High incremental ROIC (change in NOPAT ÷ change in invested capital)	Needs ever more capital per dollar of profit
Per share	EPS and FCF per share rising	Rising share count (dilution)
Diversified	Spread across customers, products and regions	One customer or product dominates
Repeatable	Recurring revenue, contracts, subscriptions	One-off orders or a cyclical peak
Not fully priced	The price implies less growth than you expect	The price assumes perfection (reverse DCF, Module 6)

5.4 What drives growth

- **Revenue = volume × price × mix.** Price-led growth shows pricing power unless it merely passes on inflation; volume-led growth shows real demand.
- **Customer businesses:** customers × retention × revenue per customer.
- **Market growth plus share gains:** a company gaining share in a market growing 3% can grow much faster than 3%.
- **Market sizing:** total addressable market (TAM), serviceable available market (SAM) and serviceable obtainable market (SOM). Be skeptical of giant TAM slides.
- **The adoption S-curve:** slow start, rapid growth, saturation. Mature companies grow roughly with the economy.
- **Base rates:** very few companies sustain growth above 20% a year for a decade. High growth fades toward the economy's rate, so your forecasts should fade too.

5.5 Sustainable and fundamental growth

$$g_{\text{sustainable}} = ROE \times b \quad g_{\text{NOPAT}} = \text{Reinvestment rate} \times ROIC$$

- **b** is the retention ratio, 1 – payout ratio.
- **Example:** ROE of 15% with a 40% payout leaves 60% retained, so the company can grow about 9% a year without new equity or more leverage.
- **Example:** a company reinvesting half its after-tax operating profit at a 12% ROIC grows operating profit about 6% a year.
- **The implication:** a company promising 15% growth with an 8% ROE and a 50% payout (sustainable growth of 4%) must borrow, issue shares or raise its returns. Ask which.

5.6 When growth creates value, and when it destroys it

$$Value = \frac{NOPAT \times \left(1 - \frac{g}{ROIC}\right)}{WACC - g}$$

Take a company with \$100 million of NOPAT, an 8% WACC and 4% growth. With no growth it would be worth \$100M ÷ 8% = \$1,250M.

Return on new investment (ROIC)	Value	Effect of growing 4% a year
15%	\$1,833M	Adds \$583M
8%	\$1,250M	Adds nothing
6%	\$833M	Destroys \$417M

Growth helps only when ROIC exceeds WACC. A company earning below its cost of capital becomes less valuable the faster it grows. Economic profit, (ROIC – WACC) × invested capital, tracks this year by year.

5.7 Forecasting growth

- **Top-down:** market size × market share. **Bottom-up:** units × price, stores × sales per store, or customers × spend per customer.
- **Consensus estimates** are a reference point, not an answer. Explain where and why you differ.
- **Three cases:** build bear, base and bull forecasts, and fade growth toward about 4% nominal (roughly long-run nominal GDP growth) by year 10.
- **Sanity checks:** the implied market share, implied margins vs the best peers, and the capital the forecast requires.

5.8 Analyzing growth in ETFs, bonds, TIPS and other assets

Asset	What "growth" means	How to analyze it
Stock ETF	Growth in the earnings and dividends of its holdings, plus valuation change	Weighted EPS growth, dividend growth, P/E vs history, total return vs benchmark (Module 8)
Dividend ETF	Growth of the income stream	5- and 10-year dividend growth, payout ratios of holdings, yield vs history
Bond or bond fund	No growth: the return is close to the starting yield over roughly its duration	Yield to maturity or SEC yield, duration, credit quality
TIPS	Principal grows with CPI; purchasing power grows at the real yield	Real yield at purchase, breakeven inflation, duration (Module 7)
REIT	Growth in rents and cash flow	FFO and AFFO per share growth, same-store NOI growth, occupancy
Gold and commodities	No cash flows, so price change only	Real interest rates, the dollar, supply and demand, the futures curve
Fund assets under management	Popularity, not performance	Signals liquidity and the fund's survival, not future returns

5.9 Step-by-step growth analysis checklist

1. Collect 5–10 years of revenue, gross profit, EBIT, EPS, free cash flow, dividends and share count.
2. Compute year-over-year growth and CAGRs for each line.
3. Split revenue growth into organic, acquired and currency effects.
4. Decompose it into volume, price and mix, or customers and revenue per customer.
5. Check margins: is operating leverage helping or hurting?
6. Check cash: free-cash-flow conversion and working-capital trends.
7. Compute ROIC and incremental ROIC, and compare them with WACC.
8. Check per-share growth and dilution.
9. Compare with peers and with industry growth.
10. Forecast three cases, fading growth over time.
11. Run a reverse DCF to see what growth the price implies (Module 6).
12. Conclude: is the growth high-quality, durable and not yet priced in?

Apply it to your case

- Show a 5-year growth table like Harbor Foods' for each stock in your report's appendix.
- For a conservative client, steady mid-single-digit growth with rising dividends and strong cash conversion beats fast but fragile growth. Say so explicitly in the IPS and in each stock thesis.

Module 6 · Equity valuation

A company's value is the present value of the cash it will generate; its price is what the market charges today. Buy when your estimate of value sits comfortably above the price, and show judges which assumptions drive the gap.

6.1 The four approaches

Approach	Methods	Best for	Weakness
Intrinsic	Discounted cash flow (DCF), dividend discount model (DDM), residual income	Predictable cash flows	Very sensitive to the discount rate and terminal growth
Relative	P/E, EV/EBITDA, EV/sales, P/B, P/FCF, PEG, dividend yield	Quick cross-checks against peers	Inherits any mispricing of the peer group
Asset-based	Net asset value, liquidation value, sum of the parts	REITs, holding companies, conglomerates, distressed firms	Misses the value of a strong ongoing franchise
Market-implied	Reverse DCF	Testing what the current price assumes	Needs judgment about what is plausible

6.2 Discounted cash flow, step by step

1. Forecast revenue, operating margin, taxes and reinvestment for 5–10 years.
2. Compute free cash flow to the firm (FCFF) for each year.
3. Discount each year's FCFF at the weighted average cost of capital (WACC, section 6.3).
4. Add a terminal value for all years after the forecast. Keep terminal growth at or below long-run nominal GDP growth; 2–3% is typical, and a rate close to WACC makes the value explode.
5. Enterprise value (EV) = present value of forecast FCFF + present value of the terminal value.
6. Equity value = EV – net debt – preferred stock – minority interests. Divide by diluted shares for value per share.
7. Run sensitivities on WACC and terminal growth, then compare with the price.

$$FCFF = EBIT(1 - t) + D\&A - \text{Capex} - \Delta NWC \quad TV_n = \frac{FCFF_n(1 + g)}{WACC - g}$$

Worked example. A company will generate FCFF of \$100M, \$108M, \$116M, \$124M and \$132M over five years. WACC is 8%, terminal growth 2.5%, net debt \$300M and diluted shares 50M.

Year	1	2	3	4	5
FCFF (\$M)	100	108	116	124	132
Discount factor at 8%	0.9259	0.8573	0.7938	0.7350	0.6806
Present value (\$M)	92.6	92.6	92.1	91.1	89.8

- Present value of the five years of cash flow: \$458.3M.
- Terminal value: $132 \times 1.025 \div (0.08 - 0.025) = \$2,460.0\text{M}$, worth \$1,674.2M today.

- Enterprise value \$2,132.5M – net debt \$300M = equity value \$1,832.5M, or **\$36.65 per share**.
- The terminal value is 78.5% of enterprise value. That share is typical, and it is why long-run assumptions dominate any DCF.

Sensitivity of value per share

WACC \ terminal growth	2.0%	2.5%	3.0%
7%	\$41.82	\$46.30	\$51.89
8%	\$33.71	\$36.65	\$40.18
9%	\$27.92	\$29.97	\$32.37

One point of WACC moves the value by roughly 20–25%. Present a range, not a single number.

6.3 The cost of capital

$$r_e = r_f + \beta \times ERP \quad WACC = \frac{E}{V}r_e + \frac{D}{V}r_d(1 - t)$$

- **Risk-free rate:** the 10-year Treasury yield, matched to the long horizon of the cash flows.
- **Beta:** a regression of the stock's returns on the market's (5 years of monthly or 2 years of weekly data), or a data provider's figure. Adjusted beta = 0.67 × raw beta + 0.33 pulls estimates toward 1.
- **Equity risk premium (ERP):** the extra return investors demand for owning stocks. Damodaran publishes an implied US ERP every month.
- **Cost of debt:** the yield to maturity on the company's bonds, or the risk-free rate plus a spread for its credit rating, taken after tax (the US federal corporate rate is 21%).
- **Weights:** use market values, not book values.
- **Example:** a 4.2% risk-free rate, a beta of 1.1 and a 5% ERP give a 9.7% cost of equity. With 80% equity, 20% debt costing 5.5% and a 21% tax rate, WACC = 8.63%.

6.4 The dividend discount model

$$P_0 = \frac{D_1}{r - g}$$

- **Example:** next year's dividend is \$2.00, the required return 8% and growth 4%: value = \$50.00. At 5% growth it is \$66.67; at 3% it is \$40.00. One point of growth moves the value by 20–33%, so the model is only as good as its growth input.
- **Where it fits:** mature dividend payers such as utilities, consumer staples, banks and REITs. Use a two-stage version when today's high growth will slow, or the H-model when it fades steadily.

6.5 Relative valuation with multiples

Pick 4–8 genuinely comparable peers (same industry, similar growth, margins, size and risk). Compute their multiples on the same basis, trailing or forward, and take the median. Apply it to your company's metric, adjust for differences in growth and quality, and cross-check with a DCF.

Multiple	Best for	Caution
P/E	Mature, profitable companies	Distorted by leverage, one-off items and buybacks; peak-cycle earnings make cyclicals look cheap
EV/EBITDA	Firms with different debt levels; capital-intensive industries	Ignores capex and taxes
EV/sales	Unprofitable or early-stage companies	Ignores profitability entirely
P/B	Banks and insurers	Meaningless for asset-light firms
P/FFO or P/AFFO	REITs	Definitions vary between REITs
FCF yield (FCF ÷ market cap)	Cash-generative businesses	Lumpy capex distorts single years
PEG (P/E ÷ growth rate)	Comparing growth companies	Growth estimates are uncertain
Dividend yield	Income stocks	A very high yield can signal a coming dividend cut

Justified multiples link a multiple to fundamentals, so you can say whether a multiple is deserved:

$$\text{Forward } P/E = \frac{\text{Payout ratio}}{r - g} \quad P/B = \frac{ROE - g}{r - g}$$

With a 50% payout, an 8% required return and 4% growth, the justified forward P/E is 12.5×; a stock at 25× needs much faster growth or much lower risk to deserve it. With a 12% ROE the justified P/B is 2.0×, and a company whose ROE merely equals its cost of equity deserves to trade at book value.

6.6 Reverse DCF: what is the market assuming?

Start from the price and solve for the growth or margins it implies, then ask whether that is believable given the company's history, its industry and its competition. In a spreadsheet, use Goal Seek to set value per share equal to today's price by changing the growth rate.

The dividend model gives a quick version: implied growth = $r - D1 \div P0$. A \$60 stock paying a \$2 dividend with an 8% required return implies 4.7% dividend growth forever. That is above long-run nominal GDP growth, so the price looks demanding. Reverse DCFs turn an argument about a number into an argument about the business, which judges find persuasive.

6.7 Other methods

- **Residual income:** value = book value today + present value of future (ROE – cost of equity) × book value. Useful for banks and companies that pay no dividend.

- **Sum of the parts:** value each segment with its own method or multiple, then subtract net debt and unallocated corporate costs. Useful for conglomerates.
- **Net asset value:** the market value of assets minus liabilities. Useful for REITs and holding companies.

6.8 Margin of safety and presenting value

- **Margin of safety** = $(\text{intrinsic value} - \text{price}) \div \text{intrinsic value}$. As a rule of thumb, ask for 25–40% on uncertain businesses and 10–20% on stable ones.
- **Football field:** a chart of horizontal bars showing the value range from each method (DCF, comparables, DDM, 52-week trading range), with the current price as a vertical line. It is the standard valuation exhibit in a report.
- **Upside and downside:** state the upside to your base case and the downside to your bear case. For a conservative client, the downside matters more.

6.9 Common valuation mistakes

- Terminal growth above long-run nominal GDP, or close to WACC.
- Mismatching cash flows and discount rates: FCFF with the cost of equity, or nominal cash flows with real rates.
- Ignoring stock-based compensation and dilution.
- Double-counting growth with both high forecast growth and a high exit multiple.
- Valuing cyclical companies on peak earnings.
- False precision: \$36.65 is a point inside a range of roughly \$30 to \$46.
- Anchoring on today's price or on analysts' targets.

Apply it to your case

- Value every stock with at least two methods (a DCF or DDM plus multiples) and include a sensitivity table.
- For a conservative client, favor businesses where even the bear case sits close to today's price. Limited downside is the point.

Module 7 • Fixed income and TIPS

Bonds are the ballast of a conservative portfolio. Know exactly what can go wrong (rates, inflation, credit, liquidity), match maturities to the client's spending needs, and use TIPS to protect purchasing power.

7.1 Bond basics

$$P = \sum_{t=1}^n \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}$$

- **Terms:** face (par) value F , coupon C , maturity n , price P and yield y .
- **Price and yield move in opposite directions.** When market rates rise, existing bonds with lower coupons are worth less.
- **Premium and discount:** a bond trades above par when its coupon exceeds market yields, and below par when it is lower.
- **Yield measures:** current yield = annual coupon $\div$ price. Yield to maturity (YTM) is the discount rate that makes the present value of all cash flows equal the price; it is your return if you hold to maturity, nothing defaults and coupons are reinvested at the same yield. Yield to worst is the lowest of the YTM and the yields to each call date. Bond funds report a standardized SEC 30-day yield.
- **Clean and dirty price:** the dirty (invoice) price adds accrued interest to the quoted clean price.
- **Spot and forward rates:** spot rates are yields on zero-coupon bonds; forward rates are the future rates implied by today's curve.

7.2 Interest-rate risk: duration and convexity

$$\frac{\Delta P}{P} \approx -D_{mod} \Delta y + \frac{1}{2} C (\Delta y)^2$$

- **Macaulay duration:** the weighted-average time, in years, until you receive the bond's cash flows.
- **Modified duration:** the approximate percentage price change for a 1-point change in yield.
- **Effective duration:** used for bonds with embedded options (callables, mortgages), measured by shocking yields up and down.
- **Convexity:** the curvature of the price-yield relationship. For ordinary bonds, prices rise more when yields fall than they drop when yields rise by the same amount.
- **DV01:** the dollar change in value for a 0.01-point (1 basis point) change in yield.

Worked example. A 10-year bond with a 4% annual coupon is priced to yield 5%.

- Price: \$922.78 per \$1,000 of face value, a discount because the coupon is below the yield.
- Macaulay duration 8.36 years, modified duration 7.96, convexity 78.3.

- If yields rise to 6%, the price falls to \$852.80, down 7.58%. Duration alone predicts -7.96%; adding convexity gives -7.57%, almost exact.
- If yields fall to 4%, the price rises to \$1,000.00, up 8.37%. The gain beats the loss for the same move: that is convexity.

Longer maturities and lower coupons mean higher duration, and a zero-coupon bond's duration equals its maturity. Rising rates hurt prices but let you reinvest coupons at higher yields; at roughly the duration horizon the two effects offset. That is the idea behind immunization: match the portfolio's duration to when the client needs the money.

7.3 Treasuries and yield-curve strategies

- **Bills** mature in 4, 6, 8, 13, 17, 26 or 52 weeks and are sold at a discount. **Notes** run 2, 3, 5, 7 or 10 years; **bonds** 20 or 30 years; **TIPS** 5, 10 or 30 years; floating-rate notes 2 years. STRIPS are zero-coupon pieces of Treasuries.
- Treasury interest is exempt from state and local income tax.
- **Ladder**: equal amounts maturing every year (for example years 1–5). It gives steady liquidity and averages out interest rates. It is the classic tool for a conservative client with known cash needs.
- **Bullet**: maturities clustered around one date, matched to a goal such as a down payment in year 3.
- **Barbell**: short and long maturities with nothing in the middle, for liquidity plus yield.
- **Riding the yield curve**: buy on the steep part of the curve; as the bond ages, its yield falls and its price rises.
- **Duration positioning**: shorten duration if you expect rates to rise, lengthen it if you expect them to fall.

7.4 Credit risk

Grade	S&P and Fitch	Moody's	Meaning
Investment grade	AAA, AA, A, BBB	Aaa, Aa, A, Baa	Strong to adequate capacity to repay
High yield	BB, B	Ba, B	Speculative
Distressed	CCC to C, and D for default	Caa to C	Vulnerable or in default

- **The cut-off**: BBB– (Baa3) is the lowest investment-grade rating. Bonds downgraded below it are "fallen angels" and often must be sold by funds restricted to investment grade.
- **Expected loss** = probability of default × loss given default (1 – recovery rate) × amount exposed.
- **Credit spread**: corporate yield minus the Treasury yield of the same maturity; the option-adjusted spread (OAS) handles bonds with options. Spreads widen in recessions, which is why corporate bonds fall when stocks fall.
- **Seniority**: senior secured debt is repaid first, then senior unsecured, subordinated debt, preferred stock and finally common equity.

- **Credit analysis:** capacity to pay (cash flow, interest coverage, debt to EBITDA), collateral, covenants and the character of management.

7.5 Other bond sectors

Sector	What it is	Appeal	Main risks
Municipal bonds	State and local government debt	Interest free of federal tax, often of state tax for in-state buyers	Credit (especially revenue bonds), rates; pointless in tax-advantaged accounts
Agency mortgage-backed securities	Pools of home loans guaranteed by Ginnie Mae, Fannie Mae or Freddie Mac	Extra yield over Treasuries	Prepayments; negative convexity
Investment-grade corporates	Debt of strong companies	Spread over Treasuries	Spread widening in recessions
High-yield corporates	Debt of weaker companies	High coupons	Defaults; behave like stocks in a crisis
International and emerging-market bonds	Foreign government and corporate debt	Diversification; higher yields in emerging markets	Currency and political risk
CDs	Bank deposits with fixed terms	FDIC insurance up to \$250,000 per depositor, per insured bank, per ownership category	Penalties for early withdrawal
Money market funds	Pools of very short-term debt	Daily liquidity	Yield falls quickly when the Fed cuts
Preferred stock	A hybrid of bond and stock	Fixed dividends	Long duration, credit and call risk
Convertible bonds	Bonds that can convert into stock	Stock upside with a bond floor	Complexity, credit risk

$$\text{Taxable-equivalent yield} = \frac{\text{Municipal yield}}{1 - t}$$

A 3.5% municipal bond is worth a 5.15% taxable yield to an investor in the 32% federal bracket.

7.6 TIPS: inflation protection explained

How they work

- The principal is adjusted for inflation using CPI-U (not seasonally adjusted), with a lag of about three months.
- A fixed real coupon is paid every six months on the adjusted principal, so interest payments rise with inflation.
- At maturity you receive the inflation-adjusted principal or the original par, whichever is greater. That is the deflation floor.
- TIPS are issued with 5-, 10- and 30-year maturities, sold at auction and traded on the secondary market.

$$\text{Adjusted principal} = \text{Par} \times \frac{CPI_{\text{today}}}{CPI_{\text{issue}}} \quad \text{Interest} = \text{Real coupon} \times \text{Adjusted principal}$$

Example. Buy \$10,000 of TIPS with a 1.5% real coupon. CPI rises 3% over the year, so the principal becomes \$10,300 and the year's interest is about $1.5\% \times \$10,300 = \154.50 , paid in two halves. If CPI fell 2% instead, the principal would drop to \$9,800 along the way, but you would still get at least \$10,000 back at maturity.

Real yield and breakeven inflation

- TIPS are quoted by their real yield: the return above inflation if held to maturity.
- Breakeven inflation = nominal Treasury yield – TIPS real yield, at the same maturity.
Illustration: a 10-year Treasury at 4.3% and a 10-year TIPS at 1.9% imply 2.4% breakeven. If inflation averages more than 2.4% over the decade, TIPS beat the nominal bond; if less, the nominal bond wins. Check current figures on Treasury's daily real yield curve.
- Breakevens also carry an inflation risk premium and get distorted when markets are stressed, so treat them as an imperfect forecast.

Risks people miss

- **Real-rate risk:** TIPS have duration. When real yields jumped in 2022, broad TIPS indexes lost roughly 12% even as inflation surged, while short-term TIPS lost only about 3% (approximate). TIPS protect against inflation, not against rising real rates, so match their maturity to the client's horizon.
- **Phantom income:** the yearly inflation adjustment to principal is taxable federal income even though you receive it only at maturity. Hold TIPS in tax-advantaged accounts, or use TIPS funds, which pay the adjustment out. TIPS income is exempt from state and local tax.
- **Deflation:** the floor protects only the original par at maturity. Inflation already accrued, or a price paid above par, can still be lost.
- **Liquidity:** the TIPS market is smaller than the nominal Treasury market, and trading costs widen in a crisis, as in late 2008.

How to own TIPS

- Individual TIPS at auction (through TreasuryDirect or a broker) or on the secondary market through a broker.
- **TIPS ladders:** buy TIPS maturing in each year you will need money, creating an inflation-protected income stream. A popular retirement-income tool.
- **TIPS funds and ETFs:** broad (all maturities) or short-term (0–5 years). Compare duration, expense ratio and SEC yield. Tickers to research: SCHP and TIP (broad), VTIP and STIP (short-term).

When TIPS fit: the client's goals are in real terms (living costs); you expect inflation above the breakeven rate or want insurance against it; or real yields are positive, so holding to maturity locks in a guaranteed real return.

I bonds (Series I savings bonds)

$$\text{Composite rate} = r_{\text{fixed}} + 2\pi_{\text{semi}} + r_{\text{fixed}} \times \pi_{\text{semi}}$$

- Rates reset every May and November, based on CPI-U.
- Electronic purchases are capped at \$10,000 per person per calendar year; check TreasuryDirect for current rules.
- You must hold them 12 months, and redeeming before 5 years forfeits the last 3 months of interest.
- Interest is tax-deferred until redemption and exempt from state and local tax.
- Good for an individual's safe money, but they cannot be traded in a simulator.

7.7 Bond funds vs individual bonds

Feature	Individual bonds	Bond funds and ETFs	Defined-maturity ETFs
Maturity	A fixed date when principal returns (absent default)	None: duration stays roughly constant	A fixed year, then cash is paid out
Price risk if held	Disappears at maturity	Persists	Fades as maturity approaches
Diversification	Needs many bonds for corporate credit	Hundreds of bonds	Many bonds
Costs	Dealer markups	Expense ratio, as low as about 0.03%	Expense ratio
Liquidity	Varies; poor for small lots	Daily	Daily
Best use	Ladders and specific goals	Core bond exposure	Goal-matched ladders in fund form (e.g., iShares iBonds, Invesco BulletShares)

7.8 Checklist for any bond or bond fund

1. **Yield:** YTM or SEC yield, not the distribution yield.
2. **Duration:** how much will it fall if rates rise 1 point?
3. **Credit quality:** average rating and the share below investment grade.
4. **Concentration:** by sector and issuer.
5. **Inflation exposure:** nominal or inflation-linked?
6. **Costs:** expense ratio and bid-ask spread.
7. **Liquidity:** trading volume and fund size.
8. **Taxes:** federal, state, phantom income, tax-exempt status.
9. **Embedded options:** call risk and mortgage prepayments (negative convexity).
10. **Fit:** does the maturity match when the client needs the money?

Apply it to your case

- Map each of Laura's cash needs to a T-bill or bond maturity. Money needed within about three years does not belong in long bonds or stocks.
- Set a duration target for the bond sleeve and justify it with your rate view and her horizon. For a risk-averse client, short-to-intermediate duration (roughly 2–5 years) is the usual default.
- Decide how much inflation protection she needs. Short-term TIPS answer the worry "I don't want inflation to eat my savings" directly.

Module 8 · ETFs, funds and index investing

For most private clients, low-cost index ETFs form the core of the portfolio. Your edge comes from choosing the right exposure at the lowest all-in cost and knowing exactly what each fund owns.

8.1 Fund structures

Structure	How it trades	Key features	Watch out for
ETF (exchange-traded fund)	Intraday on an exchange	Low cost, transparent holdings, tax-efficient	Bid-ask spreads; premiums or discounts in stressed markets
Mutual fund	Once a day at net asset value (NAV)	Automatic investing, many active managers	Capital-gains distributions; sales loads on some share classes
Index fund (ETF or mutual fund)	Either way	Tracks an index at minimal cost	Owns the index's flaws, such as concentration in the largest stocks
Active ETF	Intraday	Manager discretion in an ETF wrapper	Higher fees; must justify them
Closed-end fund	Intraday, often away from NAV	Fixed share count; frequently uses leverage	Discounts can widen in a sell-off
Exchange-traded note (ETN)	Intraday	Unsecured debt that promises an index return	Credit risk of the issuing bank
Money market fund	At a stable \$1 NAV (government funds)	Cash management	Yield follows the Fed down

8.2 How ETFs work

- **Creation and redemption:** large dealers called authorized participants (APs) deliver a basket of securities to the fund in exchange for new ETF shares, or return shares for the basket.
- **Arbitrage:** if the ETF trades above the value of its holdings, APs create shares and sell them; if below, they buy and redeem. That keeps the price close to NAV.
- **Tax efficiency:** redemptions happen "in kind" (securities, not cash), so the fund rarely has to sell holdings at a gain. Many broad equity ETFs go years without distributing capital gains.
- **Stress:** when the underlying bonds trade less often than the ETF, as in March 2020, bond ETFs can trade at noticeable discounts. The ETF often becomes the price-discovery venue.

8.3 ETF analysis checklist

What to check	Questions to ask	Where to find it
Objective and index	Which index? Weighted by market cap, equally, by fundamentals or by a factor? How often is it rebalanced?	Prospectus, index methodology
Holdings	How many? What share is in the top 10? Sector and country weights? Overlap with your other funds?	Issuer site, daily holdings file
Cost	Expense ratio, bid-ask spread, premium or discount to NAV	Issuer site, trading screen

What to check	Questions to ask	Where to find it
Tracking	Tracking difference (fund return minus index return) and tracking error (its volatility)	Fact sheet, Morningstar
Liquidity	Average daily volume, but above all the liquidity of the underlying securities; fund assets	Issuer site, trading data
Risk	Volatility, beta, maximum drawdown; for bond ETFs, duration and credit quality	Fact sheet, Portfolio Visualizer
Income	Distribution yield and, for bond funds, the SEC 30-day yield	Fact sheet
Tax	Capital-gains distribution history, qualified dividends, K-1 forms for commodity pools	Issuer site, tax documents
Issuer	Fund size, sponsor quality, securities-lending policy, closure risk for very small funds	Issuer site

8.4 Analyzing ETF growth and performance

- **Total return:** compare NAV return and market-price return over 1, 3, 5 and 10 years, annualized.
- **Rolling and calendar-year returns:** look at how the fund did in 2008, 2020 and 2022, not only in the trailing period that flatters it.
- **Risk-adjusted return:** Sharpe and Sortino ratios, maximum drawdown, and upside and downside capture against the benchmark (Module 15).
- **What it owns:** weighted-average P/E, earnings growth, ROE and dividend growth of the holdings. A fund's future return comes from these, not from its past chart.
- **Dividend funds:** 5- and 10-year dividend growth rate, yield vs its own history, and payout ratios of the holdings.
- **Assets and flows:** growing assets signal liquidity and survival, not future returns. Popular funds often launch at the peak of a theme.

Fees compound too. \$100,000 earning 6% a year before costs for 30 years:

Annual fee	Value after 30 years
0.05%	\$566,277
1.00%	\$432,194

The 0.95-point difference costs \$134,083, nearly a quarter of the low-cost result. Costs are the most reliable predictor of fund performance you control.

8.5 Active vs passive

- **The arithmetic of active management (William Sharpe, 1991):** before costs, the average actively managed dollar must earn the market return, because together active investors hold the market. After costs, the average active dollar must therefore trail it.
- **The evidence:** S&P's SPIVA scorecards have shown year after year that most active US large-cap funds trail the S&P 500 over 10- and 15-year periods.

- **Where active can earn its fee:** less efficient markets (small caps, emerging markets, high-yield and municipal bonds), customized risk control, tax-loss harvesting, and clients with special needs such as ESG screens.
- **Core-satellite:** hold a low-cost index core (often 70–90%) and add satellites (individual stocks, active funds, factor tilts) where you have a reason. This structure works well in a competition: the core shows discipline and the satellites show research.

8.6 Factor and smart-beta ETFs

Factor ETFs tilt toward characteristics that research links to long-run excess returns or lower risk.

Factor	What it buys	Why it might work	Fit for a conservative client
Value	Cheap stocks (low P/E, P/B)	Risk premium or investor overreaction	Neutral; can lag for a decade
Size	Small companies	Compensation for risk and illiquidity	Low; more volatile
Momentum	Recent winners	Investors underreact to news	Low; sharp reversals
Quality	Profitable, low-debt, stable companies	Investors underpay for dull reliability	High
Minimum or low volatility	The least volatile stocks	Investors overpay for lottery-like stocks	High; usually falls less in sell-offs
Dividend growth	Companies with long records of raising dividends	Discipline and quality	High

Factors can underperform for years, and crowded factors can crash. Choose a factor because it matches the client's risk profile, not because of its last three years. Tickers to research: QUAL (quality), USMV and SPLV (low volatility), VIG and DGRO (dividend growth), SCHD (dividend quality).

8.7 Specialized ETFs to handle with care

- **Leveraged and inverse ETFs** reset daily, so over time they drift away from their target multiple. If an index rises 10% and then falls 10%, it ends down 1%; a 2× fund ends down 4%, and a −1× inverse fund also ends down 1%. They are trading tools, not investments for a conservative client.
- **Thematic ETFs** (robotics, cannabis, space) are concentrated, often expensive and frequently launched near the peak of the theme.
- **Commodity futures ETFs** can lose money from rolling futures in contango, and some issue K-1 tax forms.
- **Buffered or defined-outcome ETFs** cap upside in exchange for absorbing a first slice of losses over a set period. They can suit a nervous client, but they cost more, and the protection resets with each outcome period.
- **Covered-call (option-income) ETFs** pay high distributions but give up much of the upside. Judge them on total return, not on yield.
- **Single-stock ETFs** add leverage or options to one company and carry concentrated risk.

8.8 Building blocks for a conservative portfolio

Tickers are examples to research, not recommendations. Confirm on the issuer's site the fees, holdings and whether the security is allowed in your simulator.

Exposure	Role	Tickers to research
Treasury bills (0–3 months)	Liquidity reserve, near-term goals	SGOV, BIL
Short-term Treasuries (1–3 years)	Stability with a little more yield	VGSH, SHY, SCHO
Short-term TIPS (0–5 years)	Inflation protection with low rate risk	VTIP, STIP
Broad TIPS	Longer-dated inflation protection	SCHP, TIP
US aggregate bonds	Core ballast	BND, AGG
Intermediate corporates	Extra income from investment-grade credit	VCIT
Municipal bonds	Tax-exempt income for high-bracket taxable accounts	MUB, VTEB
US total market or S&P 500	Core equity growth	VTI, ITOT, VOO, IVV
Dividend growth and quality	Steadier equity income	VIG, DGRO, SCHD, QUAL
Low-volatility equity	Smaller drawdowns	USMV, SPLV
Developed international equity	Diversification	VEA, IEFA
Emerging-market equity	Growth, higher risk	VWO, IEMG
REITs	Real-estate income	VNQ, SCHH
Gold	Crisis and inflation hedge	GLDM, IAU, GLD

8.9 Mutual fund share classes and costs

- **Loads:** front-end loads (Class A) charge when you buy; back-end loads (Class C or B) charge when you sell or over time; 12b-1 fees pay for marketing out of fund assets.
- **Prefer no-load, low-cost share classes.** Institutional and "admiral" classes are often cheaper versions of the same fund.
- **Total cost of ownership:** expense ratio + trading spread + tax drag from distributions + any advisory fee.

Apply it to your case

- For each fund you pick, fill in the 8.3 checklist and keep it in your report's appendix.
- Explain each fund's role in one sentence ("VTIP protects Laura's purchasing power over the next five years with little rate risk").
- Check the overlap between funds so the portfolio is not secretly concentrated in the same ten mega-cap stocks.

Module 9 · Alternatives and real assets

Alternatives can diversify a portfolio and hedge inflation, but they bring complexity, higher fees and often illiquidity. For a conservative client, use them sparingly and only through liquid, transparent vehicles you can explain in one sentence.

9.1 Why consider alternatives

- **Diversification:** returns that do not move with stocks and bonds, especially in the crises when both fall together.
- **Inflation protection:** real assets (property, commodities, infrastructure) tend to hold value when prices rise.
- **Income:** rents, royalties and private lending.
- **The costs:** fees, lock-ups, opaque pricing, complex taxes and manager risk. Reported returns of private assets are smoothed, which makes them look less volatile than they really are.

9.2 Real estate and REITs

- **Equity REITs** own property (apartments, warehouses, data centers, cell towers, offices, shopping centers, healthcare, self-storage). **Mortgage REITs** own property loans and are far more rate-sensitive and leveraged.
- **Rules:** a REIT must distribute at least 90% of its taxable income, which is why yields are high.
- **Metrics:** funds from operations (FFO) and adjusted FFO (AFFO) instead of EPS, net asset value (NAV), occupancy, same-store net operating income (NOI) growth and leverage.
- **Cap rate** = $\text{NOI} \div \text{property value}$. Rising interest rates push cap rates up and property values down.
- **Tax:** REIT dividends are mostly taxed as ordinary income, so they belong in tax-advantaged accounts where possible.
- **Behavior:** listed REITs act like stocks in the short run (they fell sharply in 2008 and 2020) and like property over the long run.

9.3 Commodities

- **Sectors:** energy, industrial metals, precious metals, agriculture and livestock.
- **Where the return comes from:** the change in spot prices, plus roll yield, plus interest on the cash collateral backing the futures.
- **Contango vs backwardation:** in contango, later futures cost more than near ones, so rolling them loses money over time; in backwardation, rolling earns money.
- **Role:** commodities are among the few assets that tend to rise with unexpected inflation, often because they cause it. But they pay no income, swing wildly and have had decade-long droughts.

- **Vehicles:** broad commodity ETFs such as PDBC (structured to avoid K-1 tax forms) or the equities of commodity producers.

9.4 Gold

From 1971, when gold began trading freely, to 2023, gold compounded at 7.87% a year, but with 26.4% annual volatility, a worst year of -32.6% (1981) and losses in 19 of 53 years. Its correlation with US stocks over that period was -0.20 ([Damodaran Online](#)).

- **Drivers:** real interest rates (gold tends to rise when real yields fall), the dollar, central-bank buying, and fear in crises.
- **Role:** a crisis and debasement hedge that pays no income. Allocations of 0–10% are common, and around 5% is easier to defend for a conservative client.
- **Vehicles:** physically backed ETFs such as GLDM, IAU or GLD. In the US, gains on these are taxed at the collectibles rate, up to 28% for long-term gains.

9.5 Private equity, venture capital, private credit and infrastructure

- **Structure:** limited partnerships with 7–12 year lives, capital called over time, and fees often around "2 and 20" (a 2% management fee and 20% of profits).
- **The J-curve:** early returns are negative because of fees and young investments; gains come later.
- **Measuring performance:** IRR, multiple on invested capital (MOIC), total value to paid-in (TVPI) and distributions to paid-in (DPI). DPI is the cash actually returned.
- **Access:** most funds require accredited investors (roughly \$1 million of net worth excluding the home, or about \$200,000 of annual income, \$300,000 jointly) or qualified purchasers (\$5 million of investments).
- **Fit:** rarely suitable for a conservative client, and not tradable in a simulator. Listed proxies exist (business development companies, listed asset managers, infrastructure ETFs), but they behave much like stocks.

9.6 Hedge funds and liquid alternatives

Strategy	What it does	Behavior in sell-offs
Long/short equity	Buys favored stocks and shorts disliked ones	Falls less than the market, but still falls
Market neutral	Balances longs and shorts to remove market exposure	Low correlation with stocks
Global macro	Bets on rates, currencies and commodities using economic views	Varies by manager
Managed futures (trend following)	Follows price trends across many futures markets	Often helps in long sell-offs; did well in 2022 as stocks and bonds fell together
Merger arbitrage	Buys takeover targets and captures the deal spread	Steady, with occasional losses when deals break
Event-driven and credit	Bets on restructurings and distressed debt	Hurt in credit crises

Liquid alternative ETFs and mutual funds bring these strategies in a daily-traded wrapper at lower cost. Tickers to research: DBMF and KMLM (managed futures), MNA (merger arbitrage). They are diversifiers at a 0–5% weight, not core holdings.

9.7 Digital assets

- **What they are:** cryptocurrencies such as bitcoin and ether, with no cash flows and returns from price changes only.
- **Access:** the SEC approved spot bitcoin ETFs in January 2024.
- **Risks:** extreme volatility (bitcoin has had several drawdowns of more than 70%), regulatory change, custody and fraud.
- **Fit:** generally unsuitable for a risk-averse client. If a client insists, cap it at a token amount they can afford to lose entirely, and document the request.

9.8 Private-banking products: annuities and structured notes

Private banks often offer these to conservative clients, so expect judges to ask about them.

- **Annuities:** a single-premium immediate annuity (SPIA) turns a lump sum into guaranteed lifetime income and insures against outliving your money (longevity risk). Multi-year guaranteed annuities (MYGAs) work like insurance-company CDs. Fixed indexed annuities credit part of an index's gain with a 0% floor, subject to caps. Variable annuities invest in funds and can carry total fees of 2–3% a year. Watch for surrender charges, and remember that guarantees depend on the insurer's strength.
- **Structured notes:** bank-issued notes that combine a bond with options, such as principal protection with capped upside, or a buffer against the first 10–20% of losses. Downsides: the issuing bank's credit risk (some Lehman Brothers "principal-protected" notes lost most of their value in 2008), embedded fees, poor liquidity and complexity.
- **The test:** could you explain the product's payoff, costs and worst case to the client in plain English? If not, it does not belong in a conservative portfolio.

9.9 Alternatives at a glance

Alternative	Liquidity	Cost	Correlation with stocks	Inflation hedge	Fit for a conservative client
Listed REITs	Daily	Low	Moderate to high	Partial	Small allocation for income
Broad commodities	Daily	Low to moderate	Low	Good against surprises	Small, optional
Gold ETF	Daily	Low	Low to negative	Partial	0–5% as a crisis hedge
Managed futures fund	Daily	Moderate	Low	Varies	Small diversifier
Private equity and credit	Locked up for years	High	Moderate (understated)	Partial	Rarely

Alternative	Liquidity	Cost	Correlation with stocks	Inflation hedge	Fit for a conservative client
Hedge fund (direct)	Monthly or quarterly at best	High	Low to moderate	Varies	Rarely
Bitcoin ETF	Daily	Low to moderate	Unstable	Unproven	Generally no
Annuities	Poor (surrender charges)	Low to high	Not applicable	Only with an inflation rider	Possible for lifetime income
Structured notes	Poor	Embedded	Depends on the design	No	Only if fully understood

Apply it to your case

- Decide whether any alternative earns a place by solving a specific problem in Laura's profile, such as inflation protection or crisis diversification.
- If you include one, cap its weight, explain its role in one sentence and show its effect in your stress tests.
- If you exclude alternatives, say why in the report (complexity, illiquidity, cost). Judges reward a deliberate "no" as much as a justified "yes".

Module 10 · Know your client

The portfolio can only be as good as your understanding of the client. Extract every goal, constraint and fear from the case file before you touch a single security, and put numbers on them so the IPS can be measured.

10.1 Discovery: what to pull out of the case file

Area	What to record	Feeds this IPS section
Personal	Age, family, dependents, career, life stage; health only as it affects the horizon	Background, time horizon
Income and spending	Salary and its stability, other income (pension, rent), annual spending, savings rate	Return need, liquidity
Balance sheet	Assets by type and account (taxable, 401(k), traditional or Roth IRA, 529, HSA), debts and their rates, home equity, net worth	Ability to bear risk, taxes
Goals	What, how much, when, and priority (need, want or wish)	Return objective, time horizon
Liquidity	Emergency reserve, known withdrawals with amounts and dates, possible surprises	Liquidity constraint
Taxes	Federal and state brackets, account types, unrealized gains, residency	Tax constraint
Legal and regulatory	Trusts, custodial accounts, employer trading restrictions, insider status	Legal constraint
Unique circumstances	Concentrated stock, inheritance, business ownership, ESG values, family obligations, charitable plans	Unique circumstances
Attitudes	Reactions to past losses, investment knowledge, stated fears, direct quotes	Willingness to take risk

Banker's note: Quote the client's own words in your report ("I don't want to lose money") and tie each major decision back to a quote. Judges notice when the portfolio visibly answers the client.

10.2 Goals-based planning

Separate the client's money into buckets, each with its own horizon and risk level:

1. **Safety (0–3 years):** the emergency reserve and known near-term spending, held in cash, T-bills, CDs or short Treasuries.
2. **Income and stability (3–10 years):** bond ladders, short and intermediate bonds, TIPS.
3. **Growth (10+ years):** diversified stocks to beat inflation over the long run.
4. **Legacy or aspirational (optional):** the longest horizon, so it can carry the most risk.

Clients think in mental accounts. Knowing the next three years of spending are safe makes it far easier to hold stocks through a crash. Rank goals as needs (must be funded), wants and wishes, and fund the needs with the safest assets.

10.3 Quantifying goals and the required return

- **Future cost** = today's cost $\times (1 + \text{inflation})^{\text{years}}$.
- **Required return:** the rate at which today's assets plus planned savings grow into the goal. Solve it with RATE in a spreadsheet, or with Goal Seek when there are several cash flows.
- **Example:** a client has \$250,000, saves \$10,000 a year and wants \$500,000 in 10 years. The required return is 4.24% a year. With 2.5% inflation that is only about 1.7% real, which a conservative mix can reasonably target.
- **If the required return exceeds what the client's risk tolerance allows:** save more, work longer, spend less, move the date or shrink the goal. Do not simply add risk.
- **Shortfall risk is real for the over-cautious.** A portfolio earning 3% when 4.5% is needed fails slowly but surely.

10.4 Risk profiling: willingness, ability and need

Dimension	What it means	How to assess it	Signs of low tolerance
Willingness (risk attitude)	Psychological comfort with losses	Questionnaires, behavior in 2008, 2020 and 2022, the language in the case	"Can't sleep", sold in a crash, prefers guarantees
Ability (risk capacity)	Financial capacity to absorb losses without derailing goals	Horizon, wealth relative to needs, income stability, liquidity needs, insurance, debt	Short horizon, large near-term withdrawals, unstable income, thin cushion
Need (required return)	The return needed to reach the goals	The required-return calculation	Goals already funded at low returns, so no need to take risk

- **When willingness and ability disagree,** the CFA Institute approach generally adopts the more conservative of the two. If low willingness threatens the goals, educate the client about shortfall risk rather than overriding them.
- **Put tolerance into numbers:** the largest acceptable calendar-year loss in percent and dollars, a volatility target, a maximum drawdown and a minimum probability of reaching the goal.
- **Ask in dollars:** "Your \$500,000 portfolio could fall to \$450,000 in a bad year. Would you stay invested?" Dollar framing draws more honest answers than percentages.

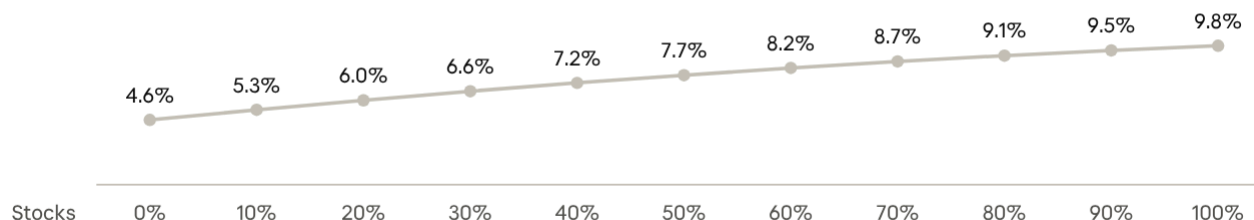
Profile	Typical share in stocks	Primary objective	Typical client
Conservative	20–35%	Preserve capital and purchasing power	Short horizon or low willingness; retirees drawing income
Moderately conservative	35–50%	Income with some growth	Near-retirees
Moderate (balanced)	50–65%	Balance growth and stability	Mid-career, medium horizon
Moderately aggressive	65–80%	Growth	Long horizon, stable income
Aggressive	80–100%	Maximum long-run growth	Young, with high capacity and willingness

These ranges are conventions, not rules. History shows what each mix of US stocks and 10-year Treasuries has actually delivered:

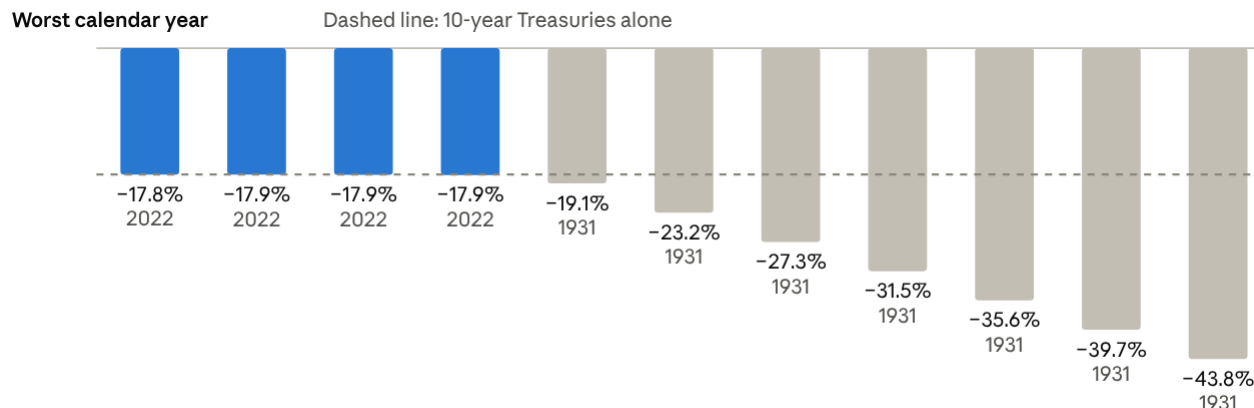
Below 40% stocks, the worst year stuck near -18%

US stocks (S&P 500) and 10-year Treasuries, rebalanced yearly, 1928–2023

Compound annual return



Worst calendar year



Computed from Damodaran Online annual returns, 1928–2023 · yearly rebalancing, no fees or taxes

Two lessons follow for a risk-averse client. Bonds cut the worst year sharply only until about 40% stocks; below that, the floor is set by the bonds themselves. And the stock-bond relationship changes: in annual data their correlation was +0.15 from 1928 to 1999 and -0.65 from 2000 to 2021, and both fell together in 1931, 1941, 1969, 2018 and 2022. Lowering the floor further takes shorter-duration bonds, T-bills and inflation protection, not just more bonds.

10.5 Human capital and total wealth

- **Human capital** is the present value of a person's future earnings. A tenured teacher's or civil servant's income is bond-like; a commissioned salesperson's or entrepreneur's is stock-like.
- **Total wealth** = financial capital + human capital. Bond-like human capital lets the financial portfolio hold more stocks; stock-like human capital, especially in the same industry as the portfolio's stocks, calls for fewer.
- **The life cycle**: young investors have large human capital and long horizons. Near retirement, human capital shrinks and the portfolio must replace the paycheck. That is the logic of target-date glide paths.
- **Do not double up**: an employee holding lots of employer stock bets both paycheck and savings on one company.

10.6 Planning foundations a private banker checks first

- **Emergency fund:** 3–6 months of expenses (6–12 months for unstable incomes or retirees), in cash and outside the investment portfolio.
- **Insurance:** health, disability, life (if there are dependents), property and umbrella liability.
- **Debt:** pay off high-interest debt such as credit cards before investing; a low-rate mortgage can stay.
- **Tax-advantaged accounts, 2026 limits** ([InvestmentNews](#), [citing the IRS](#)): \$24,500 in a 401(k), 403(b) or most 457 plans, plus an \$8,000 catch-up at 50 or older (\$11,250 at ages 60–63); \$7,500 in an IRA plus a \$1,100 catch-up at 50 or older. Roth IRA eligibility phases out between \$153,000 and \$168,000 of income for single filers and between \$242,000 and \$252,000 for married couples filing jointly.
- **Estate basics:** a will, up-to-date beneficiary designations, powers of attorney, a health-care directive, and trusts where appropriate.
- **Retirement income:** the "4% rule" (Bengen, 1994) found that withdrawing 4% of the starting portfolio, raised each year for inflation, lasted at least 30 years in US history. Early losses do the most damage (sequence-of-returns risk). Social Security benefits rise about 8% for each year a claim is delayed past full retirement age, up to 70. Required minimum distributions from traditional IRAs start at 73, rising to 75 for people born in 1960 or later.

10.7 Behavioral finance in client conversations

Prospect theory (Kahneman and Tversky, 1979) shows that losses hurt roughly twice as much as equal gains feel good. Expect these biases and plan around them.

Bias	What it looks like	How to counter it
Loss aversion	Panic over paper losses; avoiding all risk	Frame decisions in long-run dollars and goal probabilities; agree a drawdown plan in advance
Recency bias	Expecting the last few years to continue	Show long history, such as the 1928–2023 record
Overconfidence	Overtrading, concentrated bets	Position limits in the IPS
Anchoring	Fixating on a purchase price or past peak	Focus on future value, not the entry price
Herding and fear of missing out	Chasing hot themes	Require a written thesis for every purchase
Disposition effect	Selling winners too early, holding losers too long	Rules-based rebalancing and written sell criteria
Mental accounting	Treating money differently by its source	Use it constructively through goal buckets
Home bias	Overweighting one's own country	Show the diversification benefit
Status quo and regret aversion	Avoiding any decision	A default rebalancing schedule
Familiarity	Owning employer stock or favorite brands	Concentration limits

10.8 Client profile template

Field	What to record
Snapshot	Name, age, family, occupation, stability of income
Investable assets	Amount by account type (taxable, tax-deferred, tax-free)
Other assets and liabilities	Home, business, debts and their interest rates
Cash flow	Annual income, spending, savings
Goals	Each with amount, date, priority, and amount in future dollars
Time horizon	Single or multi-stage, in years
Liquidity needs	Reserve, scheduled withdrawals, possible surprises
Taxes	Brackets, account types, unrealized gains
Legal and regulatory	Trusts, restrictions
Unique circumstances	Values, concentrations, family, legacy
Willingness	Low, below average, average, above average or high, with evidence
Ability	Same scale, with evidence
Need	Required return, nominal and real
Overall risk profile	The final call and why
Key quotes	The client's own words that justify the call

10.9 Reading a conservative client

- **What "doesn't like to take risks" means:** low tolerance for visible losses, a preference for stability and income, a high value on liquidity and simplicity, and strong reactions to bad news.
- **What it does not mean:** no stocks at all. Inflation, longevity and shortfall risk punish an all-cash portfolio. In this course's worked example (Appendix), a cash-and-short-bond portfolio had about a 1% chance of reaching the client's goal, against 56% for a conservative diversified mix.
- **Your job:** find the least risky portfolio that still gives a reasonable chance of meeting the goals, and show the client both risks: losing money and missing goals.

Apply it to your case

- Fill in the client profile template from the Laura Gao file, quoting her words.
- Compute her required return for each goal.
- Rate her willingness, ability and need, and state her overall risk profile with reasons.
- Write her maximum acceptable loss in dollars and percent. It becomes the risk objective in your IPS.

Module 11 · Building the Investment Policy Statement

The Investment Policy Statement (IPS) is the contract between adviser and client: what the money is for, how much risk is acceptable, and the rules for managing it. In this competition the IPS is a graded deliverable of its own, due Nov 6, 2026, and every later decision in your final report should point back to it.

11.1 What an IPS does

- Defines the client's objectives and constraints in measurable terms.
- Sets the strategic asset allocation and the rules for managing it.
- Creates discipline: in a panic, the IPS decides what happens, not emotion.
- Defines how success will be measured and when the plan will be reviewed.
- Documents the agreed approach, which protects both client and adviser. Competition teams must follow the CFA Institute's Asset Manager Code, so write it as a fiduciary would.

11.2 The standard sections

1. **Scope and purpose:** who the client is, which assets the IPS covers and why it exists.
2. **Roles and responsibilities:** client, adviser and custodian, plus how the IPS is reviewed and amended.
3. **Investment objectives:** the return objective and the risk objective.
4. **Constraints:** time horizon, taxes, liquidity, legal and regulatory factors, unique circumstances.
5. **Strategic asset allocation:** target weights, permitted ranges and benchmarks.
6. **Investment guidelines:** permitted and prohibited investments, diversification limits, quality minimums.
7. **Rebalancing policy.**
8. **Monitoring, reporting and review.**
9. **Approval:** signatures and date.

11.3 Objectives and constraints: RRTTLLU

Element	Question it answers	What to write	Example for a conservative client
Return	What return does the client need, and want?	Required return (nominal and real), desired return, income needs	"A long-run average total return of at least 4.5% a year, about 2% after inflation, with priority on preserving capital"
Risk	How much risk can, and will, the client take?	Willingness + ability = overall tolerance, with numeric limits	"Below-average tolerance; a severe but plausible year should not cost more than 10% (\$50,000)"
Time horizon	When is the money needed?	Single or multi-stage, with dates	"Three stages: 3 years to a \$50,000 down payment, 10 years to retirement, then 25+ years of withdrawals"

Element	Question it answers	What to write	Example for a conservative client
Taxes	How will taxes affect returns?	Brackets, account types, unrealized gains	"24% federal and 5% state brackets; bonds and TIPS in the IRA, tax-efficient ETFs in the taxable account"
Liquidity	What cash is needed, and when?	Reserve, scheduled withdrawals, possible surprises	"\$50,000 in T-bills for the down payment; emergency fund held outside the portfolio"
Legal and regulatory	Are there legal limits?	Trusts, employer restrictions, prudent-investor standards	"No special legal constraints; managed to a fiduciary standard"
Unique circumstances	What else shapes the portfolio?	Values, concentrations, preferences	"Excludes tobacco and weapons manufacturers"

11.4 Writing measurable objectives

Vague (loses points)	Measurable (wins points)
"Grow her wealth safely"	"Earn at least 4.5% a year on average over rolling 5-year periods, after fees, while keeping any calendar-year loss under 10% in severe-but-plausible stress tests"
"Low risk"	"Target annual volatility of 5–7%, with a 95% one-year value at risk below 8%"
"Protect against inflation"	"Hold at least 25% in inflation-protected or short-duration assets, and beat CPI by 2 points a year over 10 years"
"Keep some cash"	"Hold 10% in T-bills to fund the \$50,000 down payment due in 3 years"
"Diversified"	"No single stock above 5%, no sector above 25% of the equity sleeve, at least 8 holdings"

11.5 Strategic allocation, ranges and the policy benchmark

- **Targets and ranges:** give every asset class a target and a band, for example stocks 33% (28–38%), bonds 52% (47–57%), cash 10% (5–15%) and gold 5% (3–7%).
- **Benchmarks:** name an index for each asset class, then blend them at the target weights into a policy benchmark, for example 33% MSCI ACWI + 52% Bloomberg US Aggregate + 10% 3-month T-bill index + 5% gold.
- **What makes a good benchmark (SAMURAI):** Specified in advance, Appropriate, Measurable, Unambiguous, Reflective of current investment opinions, Accountable (the manager accepts it) and Investable.
- **Two lenses:** a relative target (beat the policy benchmark after fees) and an absolute target (for example CPI + 2%, or the required return).

11.6 Investment guidelines

- **Permitted:** for example, US-listed stocks, ETFs, mutual funds and Treasuries. Adjust to what the simulator allows.
- **Prohibited:** margin, short selling, leveraged and inverse ETFs, crypto, private placements and any excluded industries.
- **Diversification limits:** for example, no single stock above 5%, no sector above 25% of the equity sleeve, and no single non-index fund above 30%.

- **Quality:** investment-grade bonds only (average A or better) within a stated duration band, such as 2–5 years.
- **Liquidity:** every holding tradable daily.
- **Values:** any ESG or exclusion screens the client requested.

11.7 Rebalancing, monitoring and review

- **Rebalancing:** check monthly or quarterly. Rebalance when an asset class leaves its range, and at least once a year. Use new cash flows first to reduce trading and taxes.
- **Monitoring:** performance against the policy benchmark and the absolute target, risk metrics against limits, each holding against its thesis and sell criteria, and the client's circumstances.
- **Reporting:** quarterly reports and an annual review meeting.
- **Review triggers:** a change in goals, income, health, family or tax law, or a major market event. Review the IPS at least once a year regardless.

11.8 IPS template

Copy this skeleton and fill every bracket from your client profile (Module 10).

Prepared by [team name] on [date]. Next review: [date + 12 months].

1. Scope and purpose

[Who the client is, which assets this IPS covers (amount, accounts) and its purpose.]

2. Roles and responsibilities

Client: [provides information, approves the IPS, reports changes in circumstances]

Adviser: [manages the portfolio under this IPS as a fiduciary, reports quarterly]

Custodian: [holds assets, executes trades]

3. Return objective

Required return: [x.x% nominal, x.x% real] to fund [goals].

Target: [x.x% average annual total return after fees over rolling 5-year periods].

4. Risk objective

Willingness: [level, with evidence]. Ability: [level, with evidence]. Overall: [level].

Limits: [maximum calendar-year loss in stress tests, x% (\$x); volatility target x-y%; 95% VaR].

5. Constraints

Time horizon: [stages and dates]

Liquidity: [amounts and dates; reserve]

Taxes: [brackets; which assets go in which account]

Legal and regulatory: [none, or details]

Unique circumstances: [values, preferences, concentrations]

6. Strategic asset allocation

Asset class	Target	Range	Benchmark
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[asset class]	[x%]	[x-y%]	[index]
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Policy benchmark: [weights × indexes]

7. Investment guidelines

Permitted: [...]. Prohibited: [...].

Limits: [single security, sector, credit quality, duration].

8. Rebalancing policy

[Ranges, calendar rule, use of cash flows]

9. Monitoring and review

[Reports, metrics, review triggers, annual review]

10. Approval

Client: _____ Adviser: _____ Date: _____

11.9 Mistakes judges catch

- Objectives that cannot be measured.
- A return objective that contradicts the risk objective, such as 8% a year with a 5% maximum loss.
- Ignoring inflation, taxes or fees.
- No reserve for known withdrawals.
- An allocation that contradicts the IPS, such as a "conservative" client with 70% in stocks.
- No benchmark, or a benchmark that does not match the allocation.
- No rebalancing rule.
- Generic text that could describe any client.

- Case facts copied wrongly, or arithmetic errors.

Apply it to your case

- Draft Laura's IPS from the template, filling every field from your client profile.
- Check consistency: the return target must not exceed what the allocation can reasonably earn, and the stress-test losses must stay inside the risk limits.
- Ask a teammate who did not write it to try to break it: a vague phrase, a missing constraint or a number that does not tie out.

Module 12 · Portfolio theory and asset allocation

The asset allocation drives most of a diversified portfolio's risk and its swings over time. Set the mix to the client's IPS first; choosing individual securities is a second-order decision.

12.1 Why allocation comes first

- **Brinson, Hood and Beebower (1986)** found that policy allocation explained about 94% of the variation in large US pension plans' quarterly returns over time.
- **Ibbotson and Kaplan (2000)** sharpened the claim: allocation explains about 90% of a fund's ups and downs over time, about 40% of the differences between funds, and roughly 100% of the level of return.
- **The lesson:** allocation sets the risk; selection fine-tunes it. Quote the research accurately and judges will notice.

12.2 Modern portfolio theory

Harry Markowitz (1952) showed that a portfolio's expected return is the weighted average of its parts, but its risk is not: whenever correlations are below 1, combining assets reduces risk.

$$E(R_p) = \sum_i w_i E(R_i) \quad \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

Example. Put 60% in stocks (expected return 7%, volatility 16%) and 40% in bonds (4%, 6%). Expected return is 5.8% whatever the correlation, but the risk changes:

Stock-bond correlation	Portfolio volatility
+1.0 (no diversification)	12.00%
+0.2	10.35%
0.0	9.90%
-0.2	9.42%

Lower correlation gives the same return with less risk. Diversification is the closest thing to a free lunch in finance.

- **Efficient frontier:** the set of portfolios with the highest expected return for each level of risk. Anything below it is inefficient; the minimum-variance portfolio sits at its left tip.
- **Capital allocation line:** mixing a risky portfolio with a risk-free asset. Its slope is the Sharpe ratio, and the tangency portfolio is the risky mix with the highest Sharpe ratio.
- **Risk aversion:** utility $U = E(r) - \frac{1}{2} \times A \times \sigma^2$, where A measures risk aversion. The best share in the risky portfolio is $y^* = (E(r) - r_f) \div (A \times \sigma^2)$. With an expected return of 7%, a risk-free rate of 4% and volatility of 16%, a client with $A = 4$ should hold about 29% in the risky portfolio, and one with $A = 2$ about 59%. A more risk-averse client sits closer to the risk-free asset.

12.3 CAPM and the security market line

$$E(R_i) = r_f + \beta_i [E(R_m) - r_f]$$

- **Beta** measures sensitivity to market moves: covariance with the market ÷ the market's variance. A beta of 1 moves with the market, below 1 is defensive and above 1 is aggressive.
- **Example:** with a 4% risk-free rate, a beta of 0.8 and a 9% expected market return, the required return is 8%.
- **Alpha** is the actual return minus the CAPM-required return. Positive alpha means the investment beat what its risk deserved.
- **The key insight:** only systematic risk earns a premium, because diversifiable risk can be removed for free.
- **Limits:** beta is unstable and one factor misses much of what drives returns, which led to multi-factor models.

12.4 Factor models

- **Fama–French three-factor model:** market, size (small minus big, SMB) and value (high minus low book-to-market, HML). Carhart added momentum. The five-factor model adds profitability (robust minus weak, RMW) and investment (conservative minus aggressive, CMA).
- **Arbitrage pricing theory (Ross, 1976):** returns driven by several macro factors such as growth, inflation, interest rates and credit.
- **How to use them:** regress a fund's returns on factor returns to see what it really owns. A "dividend" fund may turn out to be a value and low-volatility tilt. Kenneth French's online data library supplies free factor returns, and Portfolio Visualizer runs the regressions.

12.5 Capital market assumptions

Allocation needs forward-looking expected returns, volatilities and correlations, known as capital market assumptions (CMAs).

- **Cash:** today's T-bill yield, adjusted for the expected path of Fed policy.
- **High-quality bonds:** the starting yield is the best single predictor of the next 5–10 years' return. For credit, subtract expected default losses.
- **Stocks (Grinold–Kroner):** expected return $\approx$ dividend yield + net buyback yield + real earnings growth + inflation + yearly change in valuation. For example, 1.5% + 0.5% + 2.0% + 2.5% – 0.5% = 6.0%.
- **Volatility and correlations:** start from long history, then adjust for the regime. Stock-bond correlation turns positive when inflation is the main risk (Module 10).
- **Cite a published set** rather than inventing numbers: Vanguard, BlackRock, J.P. Morgan's Long-Term Capital Market Assumptions, Research Affiliates and GMO all publish theirs.

12.6 Strategic, tactical and dynamic allocation

- **Strategic asset allocation (SAA):** the long-term policy mix in the IPS, rebalanced back to target.
- **Tactical asset allocation (TAA):** temporary tilts inside the IPS ranges based on valuations or the cycle, for example adding 3 points to short TIPS when inflation risk rises. Each tilt needs a reason and an exit.
- **Dynamic allocation or glide path:** the mix shifts as the client ages or a goal approaches, as in target-date funds.
- **Core-satellite:** a low-cost index core with satellites of individual stocks, active funds or factor tilts.

12.7 Allocation methods

Method	How it works	Strengths	Weaknesses	Fit for a conservative client
Mean-variance optimization	Maximizes expected return for each level of risk using CMAs	Rigorous, the industry standard	Very sensitive to inputs; extreme weights	Use with constraints and sensible ranges
Black-Litterman	Starts from market-cap weights, then blends in your views by confidence	Stable, intuitive weights	More complex	Good for justified tilts
Risk parity	Each asset contributes equal risk	Balanced risk exposure	Needs leverage to reach return targets; hurt in 2022	The idea helps; the leverage does not
Equal weight (1/N)	The same weight in every asset	Simple and robust	Ignores differences in risk	Rarely
Goals-based buckets	Allocates money goal by goal, by horizon	Intuitive for clients	Can be less efficient overall	Excellent
Liability-driven investing	Matches assets to future spending (duration matching, TIPS ladders)	Funds needs with near-certainty	Lower expected return	Excellent for known spending
Rules of thumb	For example, 110 minus age in stocks	Quick	Ignores the actual client	A starting point only
Minimum variance or risk budgeting	Minimizes volatility, or sets a risk budget per asset	Low risk	Concentrates in low-volatility assets	A useful cross-check

12.8 Model allocations by risk profile

Illustrative starting points, not recommendations. Tailor every weight to the IPS.

Asset class	Conservative	Moderately conservative	Moderate	Growth
Cash and T-bills	10%	5%	5%	2%
Short-term Treasuries and TIPS	25%	20%	10%	5%
Core bonds	30%	30%	25%	15%
US stocks	20%	27%	38%	50%
International stocks	10%	13%	17%	23%
Real assets (REITs, gold)	5%	5%	5%	5%
Total in stocks	30%	40%	55%	73%

12.9 Diversification beyond asset classes

- **Geography:** the US is roughly 60% or more of world stock market value (approximate), yet investors everywhere overweight their home market.
- **Within asset classes:** sectors, factors, issuers, currencies and bond maturities.
- **Over time:** Vanguard research found that investing a lump sum beat dollar-cost averaging about two-thirds of the time, because markets rise more often than they fall. Averaging in can still help a nervous client commit.
- **Correlations rise in crises,** so match each diversifier to the kind of shock it protects against:

Shock	What has tended to protect
Growth shock (recession)	High-quality government bonds, cash
Inflation shock	T-bills, short-duration bonds, short-term TIPS, commodities, trend-following
Financial or liquidity crisis	Cash, Treasuries, the US dollar

Apply it to your case

- Choose and cite a CMA source, and put your expected returns, volatilities and correlations in a table.
- Build two or three candidate allocations within Laura's IPS ranges. Compare expected return, volatility, stress-test loss and the probability of meeting her goal (Module 14).
- Pick one and explain why it beats the alternatives for her, not in general.

Module 13 · Portfolio construction and implementation

Implementation turns the allocation into actual holdings. Good implementation is cheap, diversified and tax-aware, and every position has a written reason for being there.

13.1 From allocation to holdings

- **Top-down:** macro view → asset allocation → sectors → securities. **Bottom-up:** start from great companies. Private banks usually combine the two: top-down allocation, bottom-up selection inside each sleeve.
- **For each sleeve, choose the vehicle:** an index fund, an active fund or individual securities. Use ETFs where you have no edge (bonds, international, the broad market) and individual stocks only where research adds conviction and diversification survives.

13.2 The security selection process

1. **Screen** for candidates with quantitative filters. A conservative-friendly screen: market cap above \$10 billion, ROE above 15%, net debt below 2× EBITDA, 10+ consecutive years of dividend increases, payout ratio below 60% and beta below 1.
2. **Research** business quality, financials and growth (Modules 3–5).
3. **Value** the company with at least two methods (Module 6).
4. **Check risk:** how did it do in 2008, 2020 and 2022, and how does it correlate with what you already own?
5. **Write the thesis** (13.3).
6. **Size** the position (13.5) and set the sell criteria.
7. **Execute** with limit orders and record the trade in the log (13.9).

13.3 The one-page investment thesis

Section	What to write
Recommendation	Buy, hold or sell; target weight; role, such as "defensive dividend grower"
The business	What it does, in one sentence
Thesis	Three reasons you expect it to do well
Valuation	Base, bull and bear values per share against the price; margin of safety
Catalysts	Events that could unlock value: earnings, launches, rate changes
Risks and mitigants	The top three risks and why they are acceptable
Sell criteria	What would prove you wrong: a dividend cut, margins below a set level, valuation above a set multiple
Client fit	How the holding serves the IPS: income, stability, inflation protection

13.4 Sector analysis

The Global Industry Classification Standard (GICS) groups stocks into 11 sectors.

Sector	Cyclical or defensive	Sensitivity to interest rates	Behavior with inflation	Dividends	Metrics to watch
Information Technology	Cyclical, growth	High (long-dated profits)	Mixed	Low to moderate	Revenue growth, gross margin, R&D
Communication Services	Mixed	Moderate	Mixed	Low to moderate	Users, revenue per user, advertising
Consumer Discretionary	Cyclical	Moderate	Hurt when real incomes fall	Low to moderate	Same-store sales, margins
Consumer Staples	Defensive	Moderate (partly bond-like)	Pricing power decides	High, steady	Volume vs price, margins
Health Care	Defensive	Low to moderate	Limited	Moderate	Pipeline, patents, policy
Financials	Cyclical	Banks gain from a steeper curve	Mixed	Moderate	Net interest margin, credit quality, capital
Industrials	Cyclical	Moderate	Mixed	Moderate	Orders, backlog, margins
Energy	Cyclical	Low	Tends to benefit	High but variable	Production, breakeven price, reserves
Materials	Cyclical	Low	Tends to benefit	Moderate	Commodity prices, costs
Utilities	Defensive	High (bond-like)	Regulated pass-through, with a lag	High	Rate-base growth, allowed return
Real Estate	Mixed	High	Partial hedge	High	FFO, occupancy, cap rates

A conservative tilt usually favors consumer staples, health care, utilities and high-quality industrials. Because staples and utilities behave partly like bonds, they can lag when interest rates rise, so do not treat them as risk-free.

13.5 Position sizing

- **Equal weight** within a sleeve is simple and guards against overconfidence.
- **Conviction-weighted:** larger positions for higher conviction, within the IPS limits.
- **Risk-based (inverse volatility):** smaller weights for more volatile holdings, so each contributes similar risk.
- **IPS limits:** for example, no stock above 5% of the portfolio and no sector above 25% of the equity sleeve.
- **How many stocks?** Classic studies suggest 20–30 well-chosen stocks across sectors remove most company-specific risk in one market. With an index-fund core, a handful of individual satellites is fine.
- **Capital is not risk:** stocks can be a third of the capital and 80% of the risk (see the Appendix). Check each sleeve's contribution to total risk, not just its weight.

13.6 Trading and execution

- **Use limit orders**, especially for ETFs with wider spreads and in volatile markets. Avoid the first and last minutes of the trading day, when spreads are widest.

- **Transaction costs:** commissions (now near zero), the bid-ask spread, market impact and the cost of delay.
- **Scale in** over days or weeks if entry timing worries the client.
- **Dividend dates:** buying just before the ex-dividend date earns a taxable dividend that the price drop offsets. There is no free money.
- **In the simulator:** learn its order types, settlement and treatment of dividends and cash before you trade.

13.7 Rebalancing

- **Why:** without it, the riskiest assets grow to dominate the portfolio. Rebalancing mechanically sells high and buys low.
- **Methods:** calendar (quarterly or annual), threshold (for example ± 5 points absolute, or $\pm 20\%$ of the target) or a hybrid that checks quarterly and acts on thresholds.
- **Example:** the target is 40% stocks and 60% bonds. After a rally the mix is 48/52. On a \$500,000 portfolio, sell \$40,000 of stocks and buy bonds to return to 40/60.
- **Keep costs down:** rebalance with new contributions, dividends and withdrawals first. In taxable accounts, sell lots with losses or the highest cost basis.

13.8 Tax management (US)

- **Asset location:** hold tax-inefficient assets (taxable bonds, TIPS, REITs, high-turnover funds) in tax-deferred accounts; tax-efficient ones (broad equity ETFs, municipal bonds) in taxable accounts; and the highest-growth assets in Roth accounts.
- **Holding period:** gains on assets held more than a year are long-term and taxed at 0%, 15% or 20% depending on income, plus a 3.8% net investment income tax for high earners. Short-term gains are taxed as ordinary income.
- **Qualified dividends** are taxed at long-term rates if the holding-period test is met.
- **Tax-loss harvesting:** sell losers to realize losses that offset gains, plus up to \$3,000 of ordinary income a year. The wash-sale rule disallows the loss if you buy a substantially identical security within 30 days before or after the sale.
- **Lot selection:** specific identification (for example, highest cost first) minimizes realized gains.
- **Giving and inheritance:** donating appreciated shares to charity avoids the gain, and under current law heirs receive a step-up in cost basis.

13.9 The trade log

Judges read your trading record, and the Trading Notes Analysis is due Oct 23, 2026. Log every trade the day you make it.

Date	Ticker	Action	Shares	Price	Amount	Weight after	Reason	IPS link
Example	VTIP	Buy	280	\$50.00	\$14,000	14%	Inflation protection for needs in the next 0–5 years with little rate risk	Risk objective; inflation concern

13.10 Competition implementation tips

- Build the full portfolio early, so the IPS and final report describe a real, live portfolio.
- Do not over-trade. Every trade needs a reason tied to the IPS.
- Hold cash only where the IPS calls for it.
- Record each decision, its reason and the data behind it on the day you make it.
- Ten weeks of returns are mostly noise. Judges score the process, so manage for the client's 10-year goals, not the leaderboard.

Apply it to your case

- Write a one-page thesis for each individual stock and a one-paragraph role statement for each fund.
- Size positions within the IPS limits and check each sleeve's contribution to total risk.
- Start the trade log now, and back-fill any trades made since the competition began on September 28.

Module 14 · Risk assessment and management

For a risk-averse client, risk management is the product. Measure risk several ways, stress-test the portfolio in dollars, and show exactly how it is built to survive the scenarios the client fears.

14.1 The risk taxonomy

Risk	What it is	How to measure it	How to manage it
Equity market	Broad stock declines	Beta, volatility, drawdown	Allocation limits, diversification, quality and low-volatility tilts
Interest rate	Bond prices fall as rates rise	Duration, DV01	Duration targets, ladders, short-term bonds
Inflation	Purchasing power erodes	Real returns, sensitivity to inflation	TIPS, real assets, short duration; stocks over the long run
Credit	An issuer defaults or is downgraded	Ratings, spreads, expected loss	Investment-grade limits, diversification
Liquidity	Cannot sell quickly at a fair price	Bid-ask spreads, volume, days to sell	Daily-liquid holdings, a cash reserve
Concentration	Too much in one stock, sector or theme	Weights, top-10 share, Herfindahl index	Position and sector limits
Currency	Foreign assets lose value as the dollar rises	Foreign-currency exposure	Hedged foreign bonds; limits on unhedged exposure
Reinvestment	Maturing money must be reinvested at lower rates	Share of assets maturing soon	Ladders; some longer duration
Longevity	Outliving the money	A plan that runs to age 95 or beyond	Growth assets, annuities, Social Security timing
Sequence of returns	Bad returns early in the withdrawal years	Simulation	A cash and bond bucket for 2–5 years of spending
Shortfall	Missing the goal	Probability of success	Enough growth assets and savings
Behavioral	Panic selling at the bottom	The client's history	The IPS, education, a pre-agreed plan
Operational and counterparty	Failure of a custodian, issuer or complex product	Due diligence	Reputable custodians; avoid opaque products
Regulatory and tax	Changes in law	Monitoring	Diversify across account types
Geopolitical	Wars, sanctions, trade shocks	Scenario analysis	Diversification, Treasuries, cash, gold

14.2 Measuring risk

Measure	What it tells you	Watch out for
Standard deviation (volatility)	The typical size of swings; annualize monthly figures by multiplying by $\sqrt{12}$	Treats gains and losses alike; assumes a bell curve
Downside deviation	Volatility of returns below a target	Needs a chosen target
Beta	Sensitivity to the market	Changes over time
Maximum drawdown	The worst peak-to-trough fall	Depends on the period studied
Value at risk (VaR)	A loss not exceeded with a given confidence	Silent about how bad the tail gets

Measure	What it tells you	Watch out for
Conditional VaR (expected shortfall)	The average loss in the worst cases	Needs more data
Tracking error	Volatility of returns relative to the benchmark	Relative risk only
Duration and credit quality	Bond sensitivities	Bonds only
Probability of shortfall	The chance of missing the goal	Depends on the assumptions
Stress-test loss	The loss in a defined scenario	The choice of scenario matters

$$VaR_{95\%} = (1.645\sigma - \mu) \times \text{Portfolio value}$$

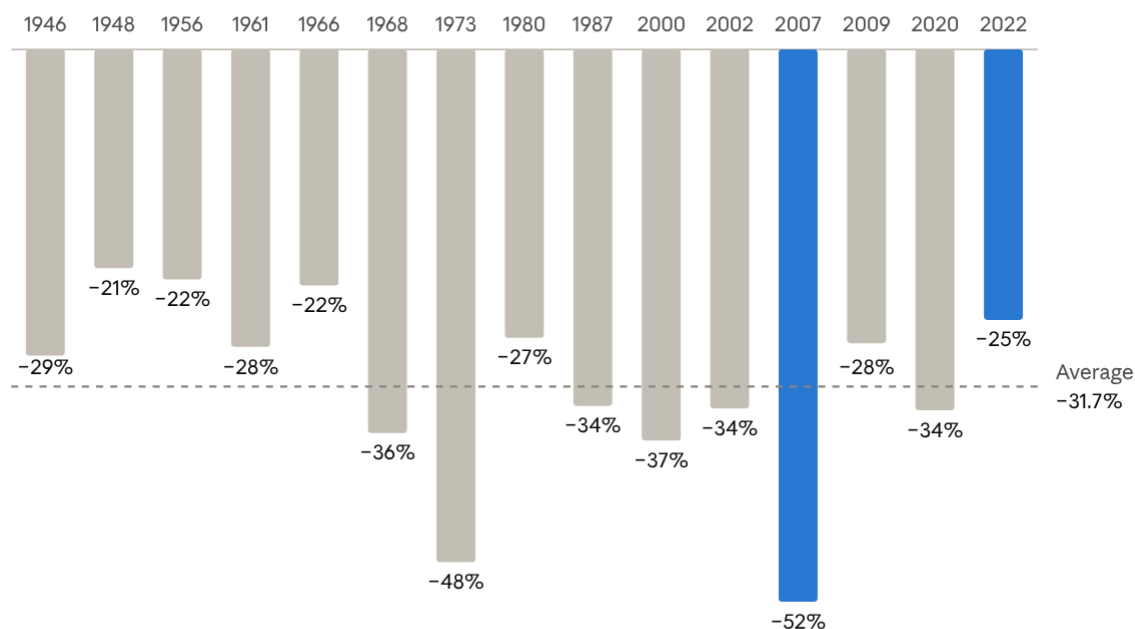
Example. A \$500,000 portfolio has an expected return of 4% and volatility of 7%. $VaR = (1.645 \times 7\% - 4\%) \times \$500,000 = \$37,575$, or 7.5%. In plain English: in about one year out of twenty, the portfolio could lose more than about \$37,600. VaR assumes a bell curve, while real returns have fat tails, and it says nothing about how bad the worst years get. Pair it with conditional VaR and stress tests.

14.3 Learning from history: bear markets

A bear market is a fall of 20% or more from a peak. From 1929 to 2022, the S&P 500 suffered 27 of them, one every 3.5 years on average, with an average decline of 35% lasting about 289 days ([Hartford Funds](#)).

US stocks have fallen 32% on average in 15 bear markets since 1946

S&P 500 decline from peak to trough, by the year each bear market began



Blue: the 2007-08 and 2022 bear markets, which the Appendix stress tests replay

Hartford Funds, S&P 500 bear markets (data as of Dec 2024) · the 15 that began after 1945

A client with a 10-year horizon will very likely live through at least one. The question is not whether stocks will fall 30%, but whether her portfolio and her nerves can handle it when they do.

14.4 Stress testing and scenario analysis

- **Historical scenarios:** replay real episodes such as the 2008 financial crisis, the 2020 COVID crash, the 2022 inflation shock, 1970s stagflation and the 1994 bond rout.
- **Hypothetical scenarios:** define shocks such as rates up 2 points, stocks down 30%, inflation at 6% for two years, credit spreads up 3 points or the dollar up 15%.
- **Method:** portfolio return $\approx$ the sum of each sleeve's weight $\times$ its scenario return. For bonds, estimate the price change as $-\text{duration} \times \text{the change in yield}$, then add a year of income.
- **Report:** show losses in dollars and percent, next to the IPS risk limit.

The scenario you choose matters. Using calendar-year returns ([Damodaran Online](#)), a portfolio of 30% S&P 500 and 70% 10-year Treasuries **gained 3.1% in 2008**, as Treasuries rallied 20.1% while stocks fell 36.6%. The same portfolio **lost 17.9% in 2022**, when both fell. A "conservative" mix can pass one crisis and fail the next, so always test both a crash and an inflation shock. The Appendix shows a full stress-test table.

14.5 Monte Carlo simulation

- **What it does:** generates thousands of possible future paths from expected returns, volatilities and correlations, then counts how often the goal is met.
- **Outputs:** the probability of success, the median outcome, the 5th and 95th percentile outcomes, and the worst drawdowns.
- **Spreadsheet version:** set each year's return to `NORM.INV(RAND(), mean, volatility)`, copy it across 10 years and down 1,000 or more rows, apply contributions and withdrawals, and count the paths that end above the goal. Portfolio Visualizer offers a ready-made Monte Carlo tool.
- **Caveats:** results are only as good as the inputs; bell curves understate fat tails; correlations shift in crises. Present ranges, not false precision.
- **Rule of thumb:** many planners aim for a 70–90% probability of success on essential goals and accept less on aspirational ones.

14.6 Risk-adjusted return at a glance

The Sharpe ratio is $(\text{return} - \text{risk-free rate}) \div \text{volatility}$: a 6% return with a 4% risk-free rate and 8% volatility gives 0.25. The Sortino ratio uses downside deviation instead, the Calmar ratio uses maximum drawdown and the Treynor ratio uses beta. Module 15 covers when to use each.

14.7 Tools to reduce and hedge risk

Tool	How it helps	Cost or drawback
Diversification	Removes company-specific risk	None
High-quality bonds and cash	Cushion stock sell-offs in recessions	Lower expected return; bonds can fail in an inflation shock
Short duration and T-bills	Limit losses when rates rise	Lower yields when rates fall
TIPS	Protect against inflation	Real-rate risk; phantom income
Gold and trend-following	Diversify crises and inflation shocks	No income; long dry spells
Defensive equity (quality, low volatility, dividend growth)	Smaller drawdowns	Can lag in strong rallies
Protective puts	Put a floor under losses	The option premium; a drag if no crash comes
Collars	Buy a put paid for by selling a call	Caps the upside
Buffered ETFs	Absorb a first slice of losses	Higher fees; capped upside
Stop-loss orders	Exit after a set fall	Gaps and whipsaws; selling after the drop
Rebalancing and position limits	Keep risk at target	Trading costs and taxes
A cash reserve for spending	Avoids forced selling in a downturn	Cash drag

14.8 Sequence-of-returns risk

When a client is withdrawing money, a bad year early does more damage than the same bad year later, because withdrawals lock in the losses. Take \$1,000,000 with \$50,000 withdrawn at the start of each year for 20 years. The returns are identical, one year of -20% and nineteen years of +7%, in a different order:

When the -20% year arrives	Value after 20 years
Year 1	\$748,786
Year 20	\$1,253,402

Without withdrawals, both orders would end at exactly the same \$2,893,222. The defense is to hold 2–5 years of withdrawals in cash and short bonds, spend from that bucket in bad years, refill it in good years, and keep spending flexible.

14.9 Managing behavioral risk

- Write the drawdown plan into the IPS: "If stocks fall 20%, we rebalance back to target; we do not sell."
- Show the client the stress tests in dollars before investing.
- Communicate early and often in bad markets: what is happening, and what the plan says to do.
- Keep near-term needs in a separate safe bucket, so the client can see they are covered.

14.10 Risk dashboard

Metric	Example IPS limit	How to compute it
Share in stocks	28–38%	Sum of equity weights
Expected portfolio volatility	5–7%	Weights × covariance matrix, or a tool such as Portfolio Visualizer
95% one-year VaR	Below 8%	$1.645 \times \text{volatility} - \text{expected return}$
Worst stress-test loss	Below 10%	Weighted scenario returns (14.4)
Bond duration	2–5 years	Weighted average of fund durations
Average credit quality	A or better	Fund fact sheets
Largest single stock	5% or less	Position weights
Largest sector	25% or less of equities	Sector weights, looking through funds
Cash and T-bills	5–15%	Sum of cash-like weights
Probability of meeting the main goal	Agreed with the client	Monte Carlo (14.5)

Apply it to your case

- Build Laura's risk dashboard and stress-test table, with losses in dollars.
- Test a 2022-style inflation shock as well as a 2008-style crash. Conservative portfolios built on long-term bonds failed 2022.
- Write her drawdown plan into the IPS.

Module 15 · Performance measurement, monitoring and client reporting

Measure performance the right way: time-weighted returns to judge the manager, money-weighted returns to show the client's own experience. Compare results with the right benchmark, explain where they came from, and report them in plain English.

15.1 Time-weighted vs money-weighted returns

- **Time-weighted return (TWR):** links the returns of each sub-period between cash flows, so the timing of the client's deposits and withdrawals does not affect it. It measures the manager's skill and is required by the Global Investment Performance Standards (GIPS) for most reporting.
- **Money-weighted return (MWR):** the internal rate of return of all cash flows. It reflects the client's actual experience, including the timing of their deposits.

$$TWR = \left[\prod_{i=1}^n (1 + r_i) \right]^{1/\text{years}} - 1$$

Example. A client starts with \$100,000. Year 1 returns +10%, taking the account to \$110,000. She then adds \$50,000, making \$160,000, and year 2 returns -5%, ending at \$152,000.

- TWR: $(1.10 \times 0.95)^{1/2} - 1 = \mathbf{2.23\% \text{ a year}}$ (4.5% cumulative).
- MWR: solve $100,000(1 + r)^2 + 50,000(1 + r) = 152,000$, giving **0.80% a year**.
- The manager earned 2.23% a year, but the client earned less because she added money just before the losing year. Both numbers are correct; they answer different questions.

15.2 Reporting conventions

- **Annualize** returns for periods longer than a year; never annualize a period shorter than a year.
- **Gross vs net:** report both before and after fees.
- **Before vs after tax, nominal vs real:** the client lives on after-tax, after-inflation returns.
- **Geometric, not arithmetic,** for multi-year averages (Module 1).
- **Same period, same basis:** compare portfolio and benchmark over identical dates.

15.3 Benchmarks

- **Policy benchmark:** the IPS target weights applied to an index for each asset class. It answers whether the implementation beat simply holding the plan.
- **Asset-class benchmarks:** each sleeve against its own index, such as US stocks against the S&P 500 or bonds against the Bloomberg US Aggregate.

- **Absolute or goal benchmark:** CPI + 2%, or the required return. It answers whether the client is on track.
- **Peer groups:** fund-category averages are useful context but suffer from survivorship bias.
- **Excess (active) return** = portfolio return – benchmark return.

15.4 Risk-adjusted performance

Measure	Formula	Use it to
Sharpe ratio	$(\text{Return} - \text{risk-free rate}) \div \text{volatility}$	Compare total-risk efficiency across portfolios
Sortino ratio	$(\text{Return} - \text{target}) \div \text{downside deviation}$	Focus on bad volatility, as a loss-averse client does
Treynor ratio	$(\text{Return} - \text{risk-free rate}) \div \text{beta}$	Judge a sleeve that sits inside a diversified portfolio
Jensen's alpha	$\text{Return} - \text{CAPM-required return}$	Measure value added after market risk
Information ratio	$\text{Active return} \div \text{tracking error}$	Judge skill relative to a benchmark
M ² (Modigliani)	The return at the benchmark's volatility	Express the Sharpe ratio in percent
Calmar ratio	$\text{Annual return} \div \text{maximum drawdown}$	Judge return per unit of worst loss
Upside and downside capture	Portfolio return $\div$ benchmark return, in up and down markets	Show a conservative client how much of the falls she avoided

Two quick readings: an active return of 0.6% with 1.5% tracking error gives an information ratio of 0.4, which is respectable. Capturing 70% of the benchmark's gains but only 50% of its losses is exactly the profile a risk-averse client wants.

15.5 Performance attribution: where did the excess return come from?

The Brinson model splits excess return into three effects:

- **Allocation:** the result of over- or underweighting asset classes: $(\text{portfolio weight} - \text{benchmark weight}) \times \text{benchmark return of that class}$.
- **Selection:** the result of picking securities within each class: $\text{benchmark weight} \times (\text{portfolio return} - \text{benchmark return})$ of that class.
- **Interaction:** the combined effect: $(\text{weight difference}) \times (\text{return difference})$.

Example. The portfolio holds 40% stocks returning 8% and 60% bonds returning 3%, for 5.0%. The benchmark holds 30% stocks returning 10% and 70% bonds returning 2%, for 4.4%. Excess return is 0.6 points.

Effect	Stocks	Bonds	Total
Allocation	+1.0	-0.2	+0.8
Selection	-0.6	+0.7	+0.1
Interaction	-0.2	-0.1	-0.3
Total	+0.2	+0.4	+0.6

Overweighting stocks in a year when they beat bonds added 0.8 points. Bond picking added 0.7, but stock picking cost 0.6. The interaction cost 0.3, because the portfolio overweighted the sleeve where its picks were weak. The Brinson–Fachler variant measures allocation against the total benchmark return; the total is the same, but the split between classes differs.

15.6 Monitoring

- **The client:** goals, income, health, family, taxes. Any change can change the IPS.
- **The markets:** the macro view and capital market assumptions, reviewed at least yearly.
- **The portfolio:** drift against IPS ranges, risk metrics against limits (Module 14 dashboard), and costs.
- **The holdings:** each thesis against its sell criteria, earnings results and news.
- **Frequency:** a monthly check of drift and risk, quarterly reporting, and an annual IPS review.

15.7 Client reporting

A good quarterly report contains:

1. Progress toward each goal, in dollars and as a probability of success.
2. Performance against the policy benchmark and the absolute target, net of fees.
3. The allocation against targets and ranges.
4. Risk metrics against the IPS limits.
5. Income received and expected.
6. Trades made and why.
7. A plain-English market commentary.
8. Next steps and decisions needed from the client.

Write for the client, not for other analysts. Explain bad quarters before the client asks, and connect every number to her goals.

15.8 In the competition

- Report the portfolio's return over the competition window against your policy benchmark over the same dates, and do not annualize a 10-week return.
- Explain the gaps with a simple attribution: allocation decisions vs security selection.
- Judges care more about whether your process was sound and true to the IPS than about the return itself. Own your mistakes and say what you learned.

Apply it to your case

- Define Laura's policy benchmark now, so you can measure against it later.
- Track the portfolio weekly in a spreadsheet: value, return, benchmark return, drift and risk metrics.
- Draft the performance section of the final report as a client letter: progress first, numbers second, jargon never.

Module 16 · Competition playbook

Winning teams tie a clear diagnosis of the client to a disciplined, well-documented strategy and defend it crisply. Wharton states that success "is not determined by portfolio performance"; teams are judged on the quality of the strategy, its fit with the client's objectives, the strength of the research and the communication ([Wharton Global Youth](#)).

16.1 How the 2026–27 competition works

All facts below come from Wharton's competition pages ([overview](#), [rules and roles](#), [FAQ](#)). The detailed instructions released to registered teams on September 15 govern formats and length limits, so check them for anything not covered here.

- **Teams:** four to six students from the same school, with a teacher advisor who may guide but may not make decisions or trade. The team leader must be at least 16.
- **Trading:** teams manage a portfolio in the Wharton Investment Simulator (WInS) from September 28 to December 4, 2026, buying and selling approved securities.
- **The case:** each year's client is a real Wharton alum with a financial scenario written for the competition.
- **Advancement:** the top 50 teams become semifinalists based on the strength of their IPS and final report. The top 10 compete at the Global Finale at Wharton on April 29–30.
- **Leaderboard:** the FAQ says WInS standings "have little to do with the final outcome."

Deliverable	Due (5:00 p.m. ET)
Official team roster	Oct 9, 2026
Trading Notes Analysis	Oct 23, 2026
Investment Policy Statement	Nov 6, 2026
Final report and school documentation	Dec 4, 2026

16.2 Rules that can disqualify you

- **Generative AI** may be used for brainstorming and idea generation, but AI-generated work may not be submitted as your own, and any AI-generated material you use must be cited. This course was drafted with an AI assistant: use it to learn, write every deliverable in your own words, and cite anything you take from it.
- **Academic integrity:** teams must follow Penn's Code of Academic Integrity. Plagiarism, fabricated analysis and misrepresenting decisions lead to disqualification.
- **Professional standards:** teams must follow the CFA Institute's Asset Manager Code.
- **No paid help:** paid advisors, education consultants and courses that teach the competition are prohibited.

16.3 The case-to-portfolio workflow

1. **Read the case three times.** First for the story, second to extract facts into the client profile (Module 10), third to collect quotes.
2. **Diagnose the client:** goals, horizon, required return, willingness, ability and need; agree an overall risk profile as a team.
3. **Form the market view:** a five-line macro outlook and a scenario table (Module 2), plus a cited set of capital market assumptions (Module 12).
4. **Draft the IPS:** objectives, constraints, allocation ranges, benchmark and guidelines (Module 11).
5. **Build and test the allocation:** compare two or three candidates on expected return, stress losses and probability of success (Modules 12 and 14).
6. **Select securities:** ETFs for each sleeve, plus individual stocks only where research earns them a place (Modules 3–8, 13).
7. **Implement and log:** trade with limit orders and record every reason in the trade log.
8. **Monitor weekly:** drift, risk dashboard, theses and news; rebalance by the IPS rules.
9. **Write the final report:** every claim traced to the client, the data or the IPS.
10. **Prepare to defend it:** presentation and Q&A drills if you reach the semifinals.

16.4 Blueprint for a conservative client

Ranges are a starting point; set the final weights from the client's own numbers.

Sleeve	Typical range	Purpose	Tickers to research
Liquidity reserve (T-bills, money market)	5–15%	Near-term needs; ready cash	SGOV, BIL
Short-term Treasuries and short TIPS	15–30%	Stability plus inflation protection	VGSH, VTIP, STIP
Core investment-grade bonds	20–35%	Income; ballast in recessions	BND, AGG
Defensive and quality US stocks	15–25%	Growth with smaller drawdowns	VIG, DGRO, USMV, QUAL
Broad US stocks	5–15%	Market growth at low cost	VTI, VOO
International stocks	5–10%	Diversification	VEA, IEFA
Real assets (gold, REITs)	0–7%	Inflation and crisis hedge	GLDM, IAU, VNQ
Individual stocks (satellites)	0–15% of the equity sleeve	Research conviction, growing dividends	Wide-moat dividend growers

Principles: fund near-term needs first; keep bond duration short to intermediate; include explicit inflation protection; size stocks to the maximum loss the client accepts; test both a 2008-style crash and a 2022-style inflation shock; prefer simple, liquid, low-cost holdings; document everything.

Traps: an all-cash portfolio that fails the goal; long-duration bonds that fail in an inflation shock; "income" traps such as junk bonds or high-yield stocks with weak balance sheets; and hidden concentration, since the S&P 500's ten largest companies now make up more than a third of the index (approximate).

16.5 The final report

Follow the official format and length limits. A structure judges can navigate:

1. **Executive summary (one page):** the client in two sentences, the objectives, the allocation, the main risks and why this portfolio fits her.
2. **Client profile:** facts, goals, quotes, risk assessment.
3. **Investment Policy Statement:** a summary or the full IPS.
4. **Market outlook:** macro view, scenarios, capital market assumptions with sources.
5. **Strategy and asset allocation:** targets, ranges and why each sleeve exists.
6. **Security selection:** the role of each fund and a short thesis for each stock.
7. **Risk management:** risk dashboard, stress tests in dollars, Monte Carlo results, drawdown plan.
8. **Performance and monitoring:** results against the policy benchmark, attribution, lessons, the monitoring and rebalancing plan.
9. **Conclusion:** how the portfolio meets each goal.
10. **Appendix:** tear sheets, trade log, calculations, data sources and citations, including any AI-generated material.

Writing rules: open each section with its conclusion. Tie each decision to a client fact or quote. Use exhibits instead of paragraphs where possible: an allocation table, a stress-test table, the risk dashboard, a valuation football field, a Monte Carlo range. Make sure every number agrees across sections. Explain any jargon in plain English the first time it appears.

16.6 The presentation

Slide	Headline (a full-sentence takeaway)
1	Our recommendation for the client in one sentence
2	Who she is and what she needs
3	Her objectives and constraints (the IPS)
4	The market outlook and what it means for her
5	The asset allocation and why each sleeve exists
6	Key holdings and their roles
7	Stress tests: what she could lose, in dollars
8	Performance so far and what we learned
9	How we will monitor and rebalance
10	Why this portfolio fits her

Rehearse to time, give every member a speaking part, and end each answer by tying it back to the client.

16.7 Judge Q&A bank

Likely question	What a strong answer covers
Why not put her entirely in Treasuries and CDs?	Required return vs the safe yield; inflation and shortfall risk; the probability-of-success comparison
How did you determine her risk tolerance?	Willingness, ability and need, each with evidence and quotes; why the more conservative view won
What is your expected return, and where does it come from?	The cited capital market assumptions and the weighted calculation; the range, not just a point
What happens if inflation returns to 5%?	Stress-test numbers; the roles of short duration, TIPS and real assets
What happens in a 2008-style crash?	Dollar loss against the IPS limit; the liquidity reserve; the rebalancing plan
Why this benchmark?	It mirrors the policy weights and meets the SAMURAI criteria
Why individual stocks instead of only ETFs?	The research edge, dividend reliability and diversification limits; or why you chose ETFs only
Why this particular fund?	Cost, tracking, liquidity, duration or holdings, and its role in one sentence
What would make you sell this stock?	The written sell criteria from the thesis
What is the biggest risk to your strategy?	Name it honestly, then give the mitigation and the warning signs you watch
How did you handle taxes?	Asset location, holding periods, loss harvesting, TIPS phantom income
How liquid is the portfolio if she needs cash tomorrow?	The share sellable within a day; the reserve sized to her known needs
Why international stocks?	Diversification, valuation, currency effects, size of the sleeve
Why gold, or why no alternatives?	Its role, its size and its stress-test behavior; or the reasons for exclusion
Your simulator return lagged. Why?	Attribution: allocation vs selection; the process held; what you would change
How does ESG factor in?	The client's stated values, or financial materiality; the effect on the benchmark
How would the portfolio change if she were 10 years older?	Horizon, human capital and withdrawal needs, leading to a lower equity share and more income
What if she wants to retire two years early?	Re-run the required return; the levers (save more, spend less, accept more risk)
How often will you rebalance, and why?	The calendar and threshold rule and its costs
What did you learn?	One specific insight about markets and one about the client

16.8 Team process

Role	Owns	Course modules
Team leader and portfolio manager	Integration, deadlines, consistency, final calls	10–12, 16
Client and IPS lead	Client profile, required return, IPS drafting	10–11
Macro and allocation analyst	Outlook, capital market assumptions, allocation, scenarios	2, 12
Equity analysts	Stock research, valuation, theses	3–6, 13
Fixed income and ETF analyst	Bonds, TIPS and fund selection	7–9
Risk and performance analyst	Stress tests, Monte Carlo, dashboard, attribution	14–15

With four to six members, some people will hold two roles. Keep a weekly rhythm (a Monday portfolio check, midweek research, a Friday decision meeting), a shared decision log and one master spreadsheet so every number in the report comes from the same place.

16.9 Data and research sources

Source	Use it for
SEC EDGAR	Company filings
FRED	Macro and market data
US Treasury	Daily nominal and real yield curves
TreasuryDirect	TIPS and I bond rules and auctions
Federal Reserve	FOMC statements, minutes, projections
Damodaran Online	Equity risk premium, industry betas and multiples, historical returns
Portfolio Visualizer	Backtests, Monte Carlo, efficient frontiers, factor regressions
Morningstar and ETF issuer sites	Fund holdings, fees, risk statistics
J.P. Morgan Guide to the Markets	Charts for the market outlook
S&P SPIVA scorecards	Evidence on active vs passive
Kenneth French Data Library	Factor returns

16.10 Mistakes that cost points

- A strategy that would suit anyone, not this client.
- Chasing the leaderboard with risky trades that contradict the IPS.
- Inflation, taxes or fees ignored.
- Numbers that disagree between sections.
- Jargon with no plain-English explanation.
- Uncited data, or AI-generated material that is neither rewritten nor cited.
- Missing a deadline: each deliverable is due at 5:00 p.m. ET.

16.11 Pre-submission checklist

-] Every goal in the case appears in the IPS with an amount and a date.
-] Return and risk objectives are measurable and consistent with each other.
-] The allocation sits inside the IPS ranges, and the benchmark matches it.
-] Stress tests cover both a crash and an inflation shock, in dollars.
-] Every holding has a stated role, and every stock has a thesis with sell criteria.
-] The trade log explains every trade.
-] Numbers agree across all sections and exhibits.
-] Every source is cited, including any AI-generated material.
-] A teammate who did not write it has proofread it.
-] The official format and length rules are met, and the file is submitted before 5:00 p.m. ET.

Appendix · Worked example: from client file to finished portfolio

This fictional case runs the whole course end to end, so your team can see what a finished analysis looks like. The client is invented and the return assumptions are illustrative; the method is what to copy. The result: a 33%-stock portfolio that expects 4.86% a year against a 4.42% requirement and keeps every stress-test loss inside the client's 10% limit.

A.1 The client file

Helena Park (fictional), 55, is a hospital administrator with a stable salary and one adult daughter.

- **Investable assets:** \$500,000, of which \$300,000 is in a taxable brokerage account and \$200,000 in a traditional IRA. A \$40,000 emergency fund sits in savings, outside the portfolio.
- **Savings:** \$12,000 a year into the taxable account.
- **Goals:** (1) give her daughter \$50,000 toward a home down payment in 3 years; (2) retire at 65 with about \$850,000, enough to draw roughly \$34,000 a year at a 4% withdrawal rate alongside Social Security and a small pension; (3) leave a gift to her local library, a wish rather than a need.
- **Taxes:** 24% federal and 5% state brackets.
- **In her words:** "I watched my 401(k) drop by a third in 2008 and couldn't sleep for months. I never want to lose more than 10% in a year." "Groceries cost a fortune now. I don't want inflation eating my savings." "Please, nothing in tobacco or weapons."

A.2 Diagnosis

$$500,000 (1 + r)^{10} + 12,000 \frac{(1 + r)^{10} - 1}{r} - 50,000 (1 + r)^7 = 850,000 \Rightarrow r = 4.42\%$$

She needs 4.42% a year, about 1.9% after 2.5% inflation.

Dimension	Assessment	Evidence
Willingness	Low	An explicit 10% loss limit; her 2008 experience
Ability	Moderate to high	10 years to retirement, stable income, an emergency fund, steady saving
Need	Moderate	A 4.42% required return
Overall	Conservative	Willingness is the binding constraint, and her ability and need do not demand more risk

A.3 IPS summary

Element	Policy
Return	At least 4.5% a year on average over rolling 5-year periods after fees (about 2% real), enough to fund the \$850,000 goal
Risk	Below-average tolerance: no calendar-year loss above 10% (\$50,000) in severe-but-plausible stress tests; expected volatility of 5–7%; 95% one-year VaR below 8%
Time horizon	Three stages: 3 years to the gift, 10 years to retirement, then 25+ years of withdrawals

Element	Policy
Liquidity	\$50,000 in T-bills in the taxable account for the gift; emergency fund kept outside the portfolio
Taxes	Bonds and TIPS in the IRA. T-bills, short Treasuries (free of state tax), tax-efficient stock funds and gold in the taxable account. No IRA withdrawals before 59½, when a 10% penalty generally applies
Legal and regulatory	None beyond fiduciary standards
Unique circumstances	Exclude tobacco and weapons makers; prefers income and simplicity
Allocation	Stocks 33% (range 28–38%), bonds and cash 62% (57–67%), gold 5% (3–7%)
Rebalancing	Check monthly; rebalance any sleeve more than 3 points from target, and the whole portfolio at least once a year, using new savings first

A.4 Capital market assumptions (illustrative)

These are teaching numbers. In your report, cite a published set (Module 12).

Sleeve	Expected return	Volatility
T-bills	3.3%	0.5%
Short-term Treasuries (1–3 years)	3.6%	1.8%
Short-term TIPS (0–5 years)	3.8%	2.5%
Core investment-grade bonds	4.6%	5.5%
US dividend-growth stocks	6.3%	14.5%
US broad market	6.5%	16.5%
International developed stocks	7.2%	17.0%
Gold	4.0%	15.0%

Correlations assumed: 0.80–0.95 among the stock sleeves, about 0.1 between stocks and bonds (–0.1 for short Treasuries), 0.6–0.8 among the bond sleeves, and 0.05–0.30 between gold and everything else.

A.5 The portfolio

Sleeve	Weight	Amount	Account	How to implement	Role
T-bills	10%	\$50,000	Taxable	SGOV or BIL	Funds the gift in year 3
Short-term Treasuries	13%	\$65,000	\$60,000 taxable, \$5,000 IRA	VGSH or SHY	Stability and liquidity; interest free of state tax
Short-term TIPS	14%	\$70,000	IRA	VTIP or STIP	Inflation protection with little rate risk
Core investment-grade bonds	25%	\$125,000	IRA	BND or AGG	Income and recession ballast
US dividend-growth stocks	13%	\$65,000	Taxable	Six individual dividend growers that pass her screen, about 2% each	Quality stocks with rising income
US broad market, screened	12%	\$60,000	Taxable	ESGV, whose index excludes tobacco and weapons makers (verify the methodology)	Low-cost market growth within her values

Sleeve	Weight	Amount	Account	How to implement	Role
International developed stocks, screened	8%	\$40,000	Taxable	A developed-markets fund screened for her exclusions	Diversification
Gold	5%	\$25,000	Taxable	GLDM or IAU	Crisis and inflation hedge

The dividend sleeve uses individual stocks because popular dividend-growth ETFs hold some companies she excludes, such as defense contractors. That turns a constraint into a chance to show stock research.

A.6 Risk check

The portfolio's expected return is 4.86% with expected volatility of 5.65%, a Sharpe ratio of 0.28 against T-bills. Its 95% one-year VaR is 4.4%, about \$22,200.

Stress tests. Scenario returns are rounded approximations of broad-index calendar-year results for 2008 and 2022, plus a hypothetical stagflation year; verify any figure before citing it.

Sleeve	Weight	2008-style crash	2022-style inflation shock	Hypothetical stagflation
T-bills	10%	+1.4%	+2.0%	+4.0%
Short-term Treasuries	13%	+6.6%	-3.8%	+1.0%
Short-term TIPS	14%	-2.5%	-2.7%	+3.0%
Core bonds	25%	+5.2%	-13.0%	-6.0%
US dividend growers	13%	-27%	-10%	-15%
US broad market	12%	-37%	-19.5%	-20%
International developed	8%	-43%	-14.5%	-22%
Gold	5%	+4.0%	+0.5%	+15%
Portfolio	100%	-9.2% (-\$46,200)	-8.7% (-\$43,500)	-5.9% (-\$29,600)
A 60/40 mix of the same sleeves		-19.0%	-12.1%	-12.2%

Every scenario stays inside her 10% limit, though the 2008-style test comes close. That is why stocks stop at 33%.

Where the risk really sits. Stocks are 33% of the money but about 82% of the expected risk; bonds and cash are 62% of the money and about 14% of the risk; gold contributes about 4%.

A.7 Will it reach the goal?

A Monte Carlo simulation ran 100,000 ten-year paths from the assumptions above, adding \$12,000 a year and withdrawing \$50,000 in year 3. The portfolio reached \$850,000 in **56%** of paths. The median outcome was \$871,000, with \$672,000 at the 5th percentile and \$1,135,000 at the 95th.

Change	Probability of reaching the goal	Worst stress-test loss
Proposed portfolio	56%	-9.2%
Raise stocks to 40%	58%	-11.9%, breaking her limit

Change	Probability of reaching the goal	Worst stress-test loss
Save \$18,000 a year instead of \$12,000	76%	-9.2%
Retire at 66 instead of 65	69%	-9.2%
Aim for \$800,000 instead of \$850,000	71%	-9.2%
All cash and short-term bonds	1%	-1.2%

For a conservative client, more risk is the weakest lever: raising stocks to 40% adds 2 points of probability and breaks her loss limit, while saving \$500 more a month adds 20 points. The adviser's job is to lay out these trade-offs and let her choose.

A.8 Monitoring plan

- **Monthly:** check drift against the ranges and update the risk dashboard.
- **Quarterly:** send the client report and review each stock against its sell criteria.
- **Annually:** rebalance fully; update the assumptions, the required return and the Monte Carlo; review the IPS.
- **Triggers for an early review:** the gift date moves, her job changes, inflation or real yields move sharply (re-check the TIPS share), or losses for the year approach 8%. Call her before she calls you.
- **The road ahead:** the T-bills fund the gift in year 3. In years 7–10, build a two-to-three-year spending bucket and consider a TIPS ladder for her first years of withdrawals.

A.9 Adapting this to the Laura Gao case

1. Replace every input with facts from the Laura Gao file: assets, savings, goals, dates, taxes and her own words.
2. Recompute her required return and her loss limit. Let them, not this example, set the weights.
3. Cite a published set of capital market assumptions, then rerun the stress tests and the Monte Carlo in a spreadsheet or Portfolio Visualizer.
4. Keep the structure: diagnosis → IPS → allocation → holdings → risk check → probability of success → monitoring.

Sources

Pages consulted for this course, with data as of the dates shown. Figures labelled approximate come from long-standing published records; verify them before citing. Research findings (Brinson, Hood and Beebower 1986; Ibbotson and Kaplan 2000; Sharpe 1991; Vanguard on lump-sum investing; S&P SPIVA) are summarized from their published conclusions, so check the originals before quoting exact figures.

- [Damodaran Online: historical returns on stocks, bonds and bills, 1928–2023](#): asset-class returns, stock/bond mix statistics, correlations and gold.
- [Hartford Funds: S&P 500 bear markets, data as of December 2024](#): bear-market declines, frequency and length.
- [InvestmentNews: IRS reveals new 401\(k\), IRA contribution limits for 2026](#): 2026 contribution limits and Roth phase-outs.
- [Wharton Global Youth: About the Investment Competition](#): format and judging.
- [Wharton Global Youth: 2026–2027 Investment Competition](#): timeline and deliverables.
- [Wharton Global Youth: Rules and Roles](#): team rules, AI policy, ethics.
- [Wharton Global Youth: Investment Competition FAQ](#): schedule and the role of WInS standings.